

DSM500 Final Project

Coursework 2 – Dissertation

Influence of Sample Size and Class Imbalance on the Optimum Level of Tabular GAN-Based Data Augmentation

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1.0 Abstract

Class imbalance presents a challenge to machine learning tasks in many domains. Minority-class scarcity biases models towards the majority class, yet the minority class is frequently the one of greatest interest. Data augmentation using generative models offers a potential way to address class imbalance in classification tasks. However, no research has systematically analysed how augmentation affects performance under controlled conditions. This study quantifies the effect of imbalance on classifier performance using tabular data and determines the extent to which synthetic data-augmentation can compensate for the consequent degradation under different conditions, relative to a counterfactual balanced scenario. To achieve this, the study implements a methodological framework that constructs scenarios of varying sample size and imbalance ratio from a common training pool and evaluates the impact of augmenting these with different amounts of synthetic data created using a generative adversarial network (GAN). The study's key finding is that simply increasing the minority class size does not guarantee improved performance when class separability is the major challenge. The study contributes to the understanding of GAN-based augmentation by highlighting several of its limitations and providing a reusable framework for evaluating imbalance, scarcity, and augmentation strategies. It concludes that further research is required to develop approaches that combine the informed sampling strategies of traditional interpolative techniques with modern generative models.

2.0 Introduction

2.1 Background

Class imbalance is a pervasive challenge in machine learning, appearing in many domains e.g. fraud detection, medical diagnosis, anomaly detection. Many classifiers degrade under class imbalance because it introduces a bias towards the majority class. This is a problem because for many applications, minority-class prediction is the key objective, and false negatives carry a high cost.

2.2 Motivation

Various approaches have been developed to address imbalance including augmentation using synthetic data. In this approach, machine learning models are trained to generate new samples from real data, which are then added to the original training set. By reducing the level of imbalance, it aims to reduce the classifier's majority bias, thereby improving performance. One type of model for generating synthetic data is CTGAN, a Generative Adversarial Network (GAN) variant specifically developed for tabular datasets. CTGAN has been researched in imbalance studies, but these typically focus on comparing different models and augmentation approaches and use different datasets. Consequently, a systematic evaluation of synthetic-data augmentation benefits under controlled conditions of class imbalance and data scarcity is not available.

2.3 Contribution

This study presents a methodology that allows imbalance ratio (IR) and sample size to be controlled independently, enabling the effects of both to be examined. Further, it allows the effects to be quantified relative to a counterfactual balanced scenario. Using this methodology, the project also examines the extent to which synthetic-data augmentation using CTGAN can recover classifier performance lost due to imbalance. To quantify the recovery, a normalised F1 recovery metric is defined which allows results to be compared across different data conditions.

2.4 Dataset

The methodology presented in this study requires a dataset of sufficient size that scenarios can be constructed simulating different imbalance ratios (IR) and sample sizes, and comprised of tabular data as would be typical of many imbalanced classification tasks. For this purpose, the Home Credit Default dataset was selected (Morelia, 2018).

2.5 Aims

To characterise the effects of class imbalance and sample size on classifier performance and to evaluate the extent to which GAN-based data augmentation can restore performance under these conditions.

2.6 Research questions

RQ1: How much does classifier performance degrade, relative to a counterfactual balanced scenario, when varying levels of data scarcity and imbalance are induced?

RQ2: Does an optimal augmentation level exist, and how does it vary across different data conditions?

RQ3: To what extent can CTGAN-based augmentation compensate for the performance degradation caused by class imbalance, relative to a balanced counterfactual?

2.7 Objectives

Objective 1: Characterise how classifier performance degrades as sample size decreases and class imbalance increases, relative to a balanced counterfactual.

Objective 2: Analyse how the optimal CTGAN-based augmentation level behaves under increasing data scarcity and class imbalance.

Objective 3: Evaluate the effectiveness of CTGAN-generated samples in recovering classifier performance relative to the balanced counterfactual across all scenarios. The balanced counterfactual provides a reference point for calculating a normalised recovery metric, enabling the true benefit of data augmentation in mitigating performance degradation caused by class imbalance to be assessed.

3.0 Literature survey

Class imbalance is a pervasive challenge in classification tasks across many domains. When the minority class is underrepresented, classifiers struggle to learn its distribution (He and Garcia, 2009; D'Souza et al., 2024) and tend towards predicting the majority class. This is particularly problematic in binary classification applications such as fraud detection, insurance claim prediction and credit-risk assessment, where the minority class is often the class of greatest interest and misclassification carries significant financial cost.

3.1 Oversampling and augmentation

In a recent survey of small-sample imbalance, Zhao et al. (2025) define imbalance as a highly skewed class distribution, while other studies find that it degrades classifier performance because the minority class is insufficiently represented for the model to learn its underlying structure (He and Garcia, 2009; D'Souza et al., 2024; Engelmann and Lessmann, 2021). Data scarcity amplifies this effect because when overall sample size is small, the minority class becomes even more weakly represented, increasing the risk of overfitting (Alzubaidi et al., 2023; Zhao et al., 2025).

Traditional oversampling methods such as SMOTE interpolate synthetic samples from real minority instances (Chawla et al., 2002) but can introduce noise and lead to overfitting (Engelmann and Lessmann, 2021). GAN-based augmentation has been proposed as a more recent alternative. GANs train a generator to learn the distribution of a population and generate synthetic samples from latent noise (Goodfellow et al., 2014). Studies show that GANs can be more effective than linear interpolation for imbalanced data (D'Souza et al., 2024). However, GANs face several challenges in replicating tabular data (Xu et al., 2019) including modelling both discrete and continuous features, non-Gaussian numerical features, and highly skewed categorical features. Accordingly, GAN-based augmentation is not guaranteed to improve performance. D'Souza et al. (2024) note that synthetic minority samples are often lower quality than majority samples, consistent with Chawla et al.'s (2002) observation that minority distributions are harder to learn. GANs are also susceptible to mode collapse, producing low-diversity samples when minority data are scarce. CTGAN, introduced by Xu et al. (2019), extends the conditional GAN framework of Mirza and Osindero (2014) and the original GAN architecture of Goodfellow et al. (2014), incorporating several features to overcome the challenges associated with tabular data. First, for multi-modal features CTGAN identifies modes and applies mode-specific normalisation and represents them as vectors suitable for neural networks. Second, training and sampling are conditioned, with categorical classes selected based on log-frequency to ensure adequate minority class representation. Other variants designed for tabular data have also improved classifier performance (Engelmann and Lessmann, 2021).

Beyond GANs, variational auto encoders (VAE) are another family of generative models introduced by Kingma and Welling (2014). A VAE is a Bayesian generative model that learns a probability distribution from data and can sample new synthetic data from that distribution. Xu et al. (2019) develop TVAE – a tabular variant of VAE. They compare the data quality of CTGAN, TVAE, and other GAN variants and find that GAN variants outperform TVAE, with CTGAN performing best overall. However, a more recent comparative study finds that SMOTE outperforms CTGAN (Kotelnikov et al, 2023).

Zhao et al. (2023) developed CTAB-GAN+ which overcomes several limitations of CTGAN including alternative normalisation for single mode continuous and high dimensional categorical variables, avoids mode collapse by using a more stable loss function (Wasserstein distance), and differentiating between continuous and discrete modes within the same variable.

Denoising diffusion probabilistic models (DDPM) are another approach to generating synthetic data. While most commonly used for computer vision tasks, Kotelnikov et al. (2023) develop a variant for tabular data – TabDDPM – and compare it to SMOTE, TVAE, CTGAN, and CTAB-GAN+. In their study TabDDPM performs best overall. Their findings reflect a broader pattern in the literature that while individual models may perform well under specific conditions, no generative model family stands out as being universally superior.

3.2 Synthetic data quality

Methods for evaluating synthetic data quality broadly fall into two categories and are used consistently in generation model research: fidelity, which measures statistical similarity and utility, which measures degradation in performance on some down-stream machine learning task (Xu et al., 2019; Zhao et al., 2023; Kotelnikov et al., 2023). Fidelity measures include pair-wise correlation difference and Kolmogorov-Smirnov statistic for numerical variables, and Jensen-Shannon divergence (JSD) and Chi-square for categorical features. Zhao (2019) finds that JSD is unstable if distributions do not overlap and also uses Wasserstein distance applied to min-max normalised values. As noted by Won et al. (2024), higher fidelity does not guarantee higher performance on machine learning tasks, and suggests performing evaluation using a model representative of one used by a practitioner.

3.3 Model performance

Won et al. (2024) more recently compare SMOTE with deep learning techniques Gaussian copula, TVAE, and CTGAN, in their performance at data augmentation. They find that although SMOTE does not achieve the best fidelity, it does provide the greatest classifier improvement through data augmentation. Fernandez (2018) explains that this is possible due to informed sampling of the SMOTE algorithm which creates samples close to the decision boundary, benefitting the classifier more than simply replicating the original distribution. Won et al. (2024) acknowledge that TVAE and CTGAN may have underperformed due to insufficient data which can harm deep learning techniques. The study has other limitations: it uses a single dataset; it evaluates only one level of augmentation; and does not perform threshold tuning which would normally be completed for an imbalanced classification application.

3.4 Identified knowledge gaps in the literature

Most studies evaluating GAN-based augmentation vary imbalance, sample size and complexity by selecting multiple datasets with differing characteristics. While this allows broad patterns to be observed, it introduces dataset-specific confounding factors that prevent precise analysis of performance behaviour (D'Souza et al., 2024; Engelmann and

Lessmann, 2021). Because sample size and imbalance vary simultaneously across datasets, their separate effects cannot be isolated.

Zhao et al. (2025) identify two unresolved questions central to understanding GAN-based augmentation. First, they ask, “how do synthetic and real data impact model performance differently?”, highlighting the absence of methods capable of manipulating real and synthetic sample sizes in a controlled manner. Second, they observe that data characteristics are the major influence on classification performance, implying that augmentation effectiveness will vary across datasets even when imbalance and sample size are held constant. This reinforces the difficulty of analysing augmentation behaviour using multiple datasets.

Research also compares augmented and un-augmented performance without considering the counterfactual balanced dataset drawn from the same population. Without this, it fails to quantify the degradation in classifier performance caused by imbalance (He and Garcia, 2009; Zhao et al., 2025) and, consequently, the proportion of performance loss recovered (He and Garcia, 2009).

Finally, while some studies vary augmentation levels, none systematically analyse augmentation benefit across different levels or identify the optimal augmentation level under varying data conditions (D’Souza et al., 2024). Most evaluate augmentation at one arbitrary level, such as the amount required to perfectly balance the training set – assuming this guarantees the optimum classifier (Won et al., 2024). But this approach ignores the risks of over- and under-fitting, meaning that performance behaviour under augmentation remains poorly understood.

3.5 Significance of addressing knowledge gaps

Because no studies directly control sample size and imbalance, it is difficult to separate their associated effects on model training from the underlying class separability in the data. This matters because addressing imbalance in the training set is not guaranteed to improve classifier performance and without this insight, practitioners risk pursuing unproductive development strategies.

No guidance exists on what levels of augmentation are likely to optimise models, or under what conditions of sample size and class imbalance. Because studies have not researched specifically whether augmentation is a significant model parameter, practitioners are more likely to assume a fixed level without exploring the parameter space.

3.6 Summary and justification for the study

Taken together, the lack of systematic evaluation of augmentation levels, and the absence of controlled studies examining sample-size and imbalance interactions highlight a clear gap in the literature. These gaps justify the need for a study that examines augmentation behaviour under controlled imbalance and varying sample sizes.

4.0 Data description

This study uses the Home Credit Default Risk dataset, released by the international consumer finance provider Home Credit for a Kaggle machine-learning competition (Morelia, 2018). The dataset is accessible through Kaggle under a competition-specific licence that permits use for academic and other non-commercial purposes. The full dataset comprises one main file and several supplementary files, with each record in the main file being linked to zero or many records in any of the supplementary files. Using feature engineering, records in the supplementary files can be aggregated and used to enrich the main file and improve classifier performance. However, this study focuses on examining the effects of imbalance, data scarcity, and data augmentation, and not training the best possible classifier, therefore only the main file was used. The data describes loans to clients, with the binary target label indicating whether the client was late in making any of the first of a defined number of instalments on the loan.

The Home Credit dataset was appropriate for this study because it provides a large tabular feature space with a mix of datatypes which will challenge GAN training. Crucially, it includes sufficient minority-class samples to support construction of experimental scenarios ranging from large and balanced to small and highly imbalanced. Finance credit default is a well-known imbalanced classification problem, and using data from this domain helps ensure that the study's findings are relevant to real-world applications.

The dataset has 307,511 records including approximately 8% positive class. The feature set comprises 120 anonymised variables including binary, continuous, and categorical types. Both categorical and numerical features contain missing values.

5.0 Methodology

This chapter outlines the methodological approach used to investigate how synthetic-data augmentation interacts with class imbalance and data scarcity in classification tasks. It describes the experimental framework developed to simulate controlled data regimes, the procedures for generating synthetic samples using CTGAN, and the evaluation strategy used to assess classifier performance. The chapter also introduces the theoretical recovery metric designed to quantify the extent to which augmentation can compensate for imbalance. Implementation details are included where they impact results or are required for reproducibility. The following overview summarises the structure of the experimental design before each component is detailed in subsequent sections.

5.1 Overview

Class imbalance in classification problems presents a persistent challenge, making it essential to understand how models behave under different data regimes, and how data augmentation strategies influence their performance. This study examines synthetic data generated using CTGAN as an augmentation strategy. It aims to determine whether an optimum level of augmentation exists, how that optimal level changes with data imbalance and scarcity, and the extent to which performance degradation due to imbalance can be recovered.

To address these research questions, an experimental framework is constructed that simulates controlled scenarios using datasets of varying sizes and imbalance ratios (IR). For each scenario, the required numbers of positive and negative samples are drawn from a larger dataset, producing size-tiered subsets in which IR varies while total sample size remains fixed (Figure 1).

For every imbalanced scenario, a GAN is trained and used to generate synthetic minority-class samples. The scenarios are then used to train classifiers using only the real data, and again using datasets augmented with different quantities of synthetic samples (Figure 2). Model performance is evaluated using cross-validation with threshold selection, followed by evaluation on a holdout test set kept separate from training data. Metrics appropriate for imbalance classification are used throughout. As an additional control, one size tier is selected for a real-data augmentation experiment in which synthetic samples are replaced with observations drawn from the training pool, and each scenario is reevaluated using the same methodology as the synthetic-data augmented scenarios.

To quantify how closely each augmented dataset approaches an idealised “balanced world” performance, a theoretical recovery metric is defined. Although this metric is anchored in a hypothetical evaluation setting, it provides diagnostic insight into the limits of naïve augmentation and the extent to which synthetic data can compensate for imbalance.

This methodology enables systematic comparison across data regimes, revealing how augmentation interacts with imbalance severity, dataset size, and classifier behaviour.

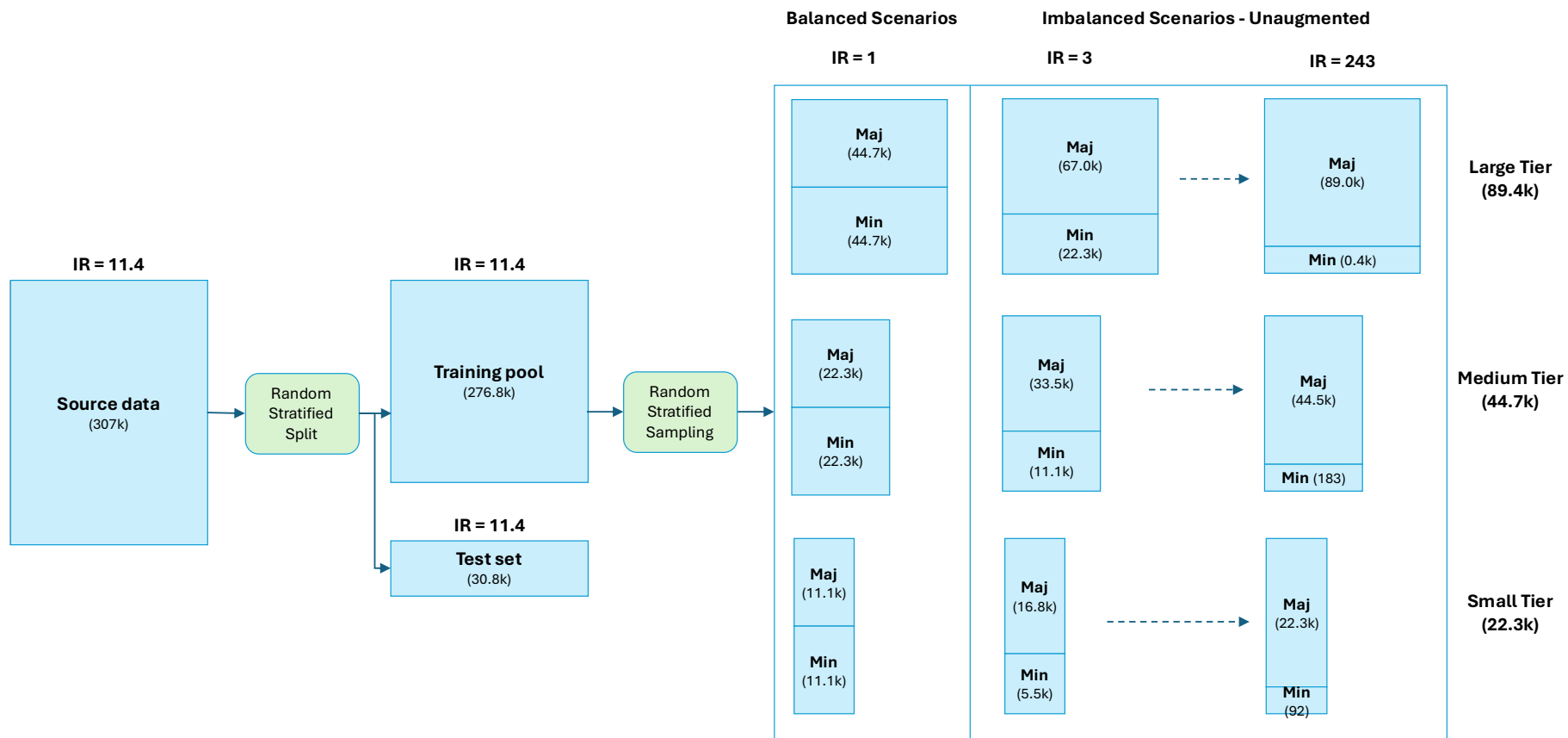


Figure 1. After creating a common stratified test set, balanced and imbalanced scenarios are created by sampling from the remaining training pool.

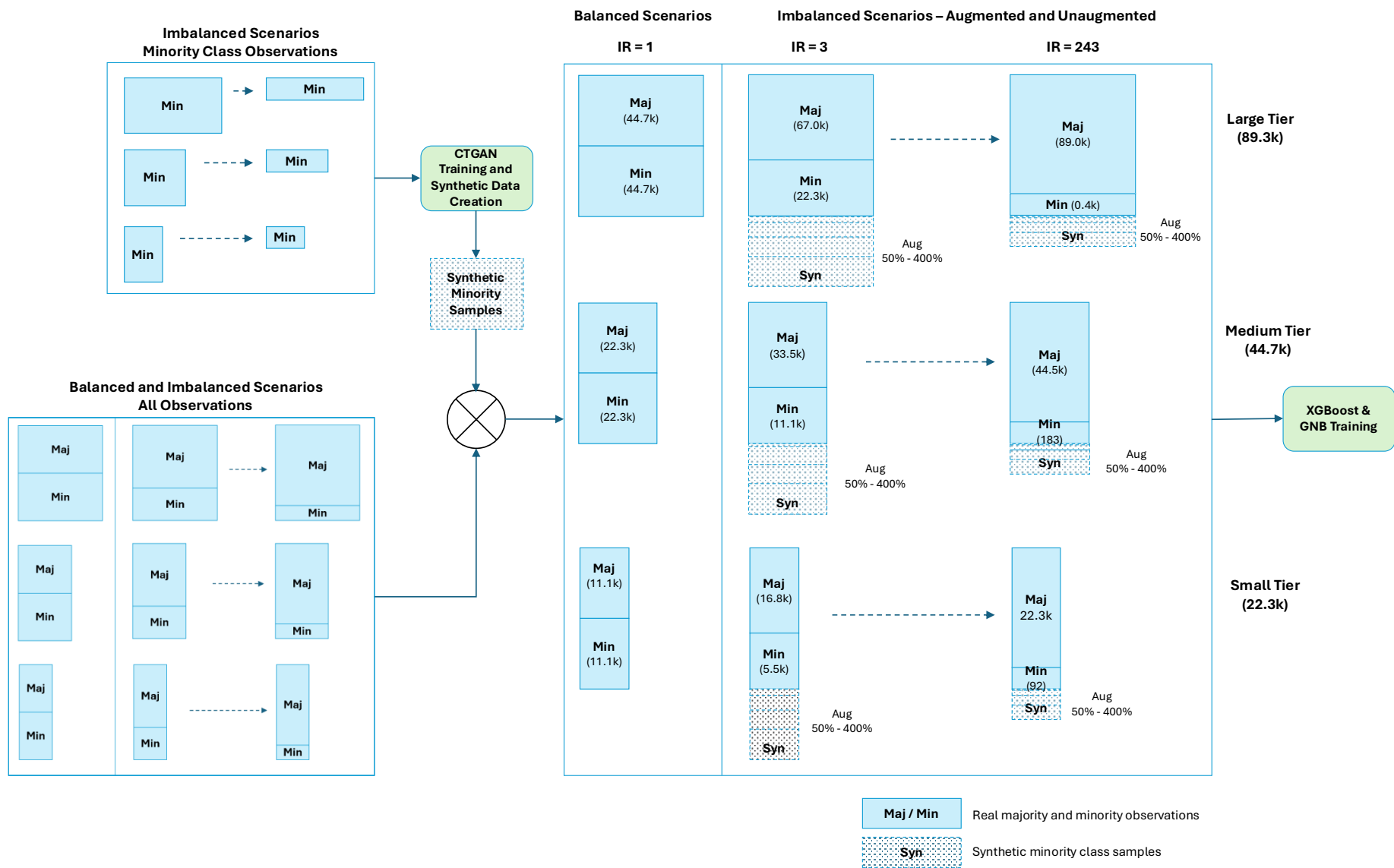


Figure 2. CTGAN is trained using minority samples from imbalanced scenarios, then used to create different levels of synthetic data to augment them.

5.2 Evaluating the effect of imbalance and sample size

This section addresses Objective 1 by constructing experimental scenarios that independently vary imbalance ratio (IR) and sample size to determine their effects on classifier performance. Scenarios are constructed using IRs and sample size which vary exponentially to provide smoother gradients for interpretation. XGBoost is used as the main classifier because it performs well with mixed-type tabular data, while Gaussian naïve Bayes is included to provide a control model which requires no hyperparameter tuning. Evaluation is performed using cross-validation and using a hold-out test which is common to all scenarios. Evaluation metrics suitable for imbalanced classification are used including precision, recall, F1-score, log-loss, PR-AUC, and ROC-AUC. Objective 1 is further addressed with the inclusion of an experiment to assess the performance of a balanced training set under the same evaluation conditions as the imbalanced scenarios.

5.2.1 Data partitioning

The available dataset of around 307k samples is partitioned 90/10 into a training pool and a holdout test set using stratified sampling to maintain the same proportions of minority and majority samples in both sets. The test set, once created, remains fixed for all scenarios and augmentation conditions. This allows every model to be evaluated against the same reference distribution regardless of its dataset size, IR, or augmentation level. Using a common test set provides a stable baseline for comparing performance across scenarios and isolates the effects of imbalance and augmentation from variation in test set composition.

The remaining data forms the training pool from which all experimental scenarios are constructed. Scenarios of different sizes and IRs are drawn exclusively from this pool, ensuring that test data is not used for classifier training, GAN fitting, or augmentation. Cross-validation is performed only within each scenario's training data, with classification thresholds selected inside the cross-validation loop to avoid leakage of information from the validation set.

This partitioning strategy supports controlled experimentation, with the test set providing a consistent external benchmark and the training pool enabling systematic construction of size-tiered and IR-varying scenarios.

5.2.2 Scenario creation

The methodology uses a primary grid of experimental scenarios, and supplementary scenarios that can be used to provide extra granularity or breadth, depending on initial results. Scenarios are constructed to vary systematically by dataset size and IR. They are organised into size tiers, within which the total number of samples is held constant while IR is varied (Table 1). This structure allows the effects of dataset size and class imbalance to be examined independently. For each scenario, positive and negative observations are sampled randomly without replacement from the training pool, then combined.

Both tier size and IR progress through logarithmic sequences: approximately 90,000, 45,000, and 22,500 for size, and 1, 3, 9, 27, 81, and 243 for IR. These geometric progressions provide smooth and interpretable gradients of data scarcity and imbalance, enabling clear observation of how classifier performance and augmentation behaviour

change as minority-class scarcity increases. The maximum sample size is constrained by the available minority samples in the training pool, while the minimum must allow a viable minority-class size for GAN training in the most extreme IR scenario. At the boundary of the grid, the large tier at IR 1 cannot be constructed because the training pool does not contain enough minority samples. Increments of three are used for IR to balance the granularity of the grid with the computation cost of executing the scenarios. While the maximum of IR 243 can be exceeded in certain domains, 243 still represents an extreme level of imbalance and provides a reasonable range for analysis.

Table 1. Scenario sample sizes are held constant within each tier as IR is varied. Insufficient minority samples are available to create a large balanced scenario.

	Large		Medium		Small	
	$N_{\text{total}} = 89,368$		$N_{\text{total}} = 44,684$		$N_{\text{total}} = 22,342$	
	N_{min}	N_{maj}	N_{min}	N_{maj}	N_{min}	N_{maj}
IR 1	-	-	22,342	22,342	11,171	11,171
IR 3	67,026	22,342	33,513	11,171	16,756	5,586
IR 9	80,431	8,937	40,216	4,468	20,108	2,234
IR 27	86,176	3,192	43,088	1,596	21,544	798
IR 81	88,278	1,090	44,139	545	22,070	272
IR 243	89,002	366	44,501	183	22,250	92

To avoid physically replicating data, only sample indices are saved, as implemented in Appendix E, Notebook 02 (Experimental Scenarios). For each scenario, one complete sample and k training and validation folds are created. Five folds are used for cross-validation because this number provides sufficient stability in performance estimates while maintaining adequate minority class representation within each fold. Using more than five folds would produce very few minority samples per fold in the extremely imbalanced scenarios and would increase the computational cost.

To ensure reproducibility and sample independence, random seeds based on sample size, number of minority samples, and bootstrap number are used.

5.2.3 Data preprocessing and missingness indicators

Because the Home Credit dataset (Morelia, 2018) is relatively clean and has been anonymised by the provider, only limited preprocessing is performed. Integers are converted to Boolean for binary-valued or categorical for multi-valued variables. Missing values for categorical values are simply assigned a “missing” category, while for numerical columns indicator features are added to record missingness and median values are assigned during training. This approach avoids skewing numerical data while retaining the informativeness of missing values. Missing indicators are added only for columns with at least 0.5% missing values to avoid introducing unnecessary features. Additionally, it was observed that the missingness of 42 of the numerical features is highly correlated (Figure 3). While not perfectly correlated, a single indicator calculated by majority vote is used to represent missingness in these features. While potentially losing some information, this approach is justified by avoiding adding many new columns which risks increasing noise and overfitting, and accordingly impairing classifier performance.

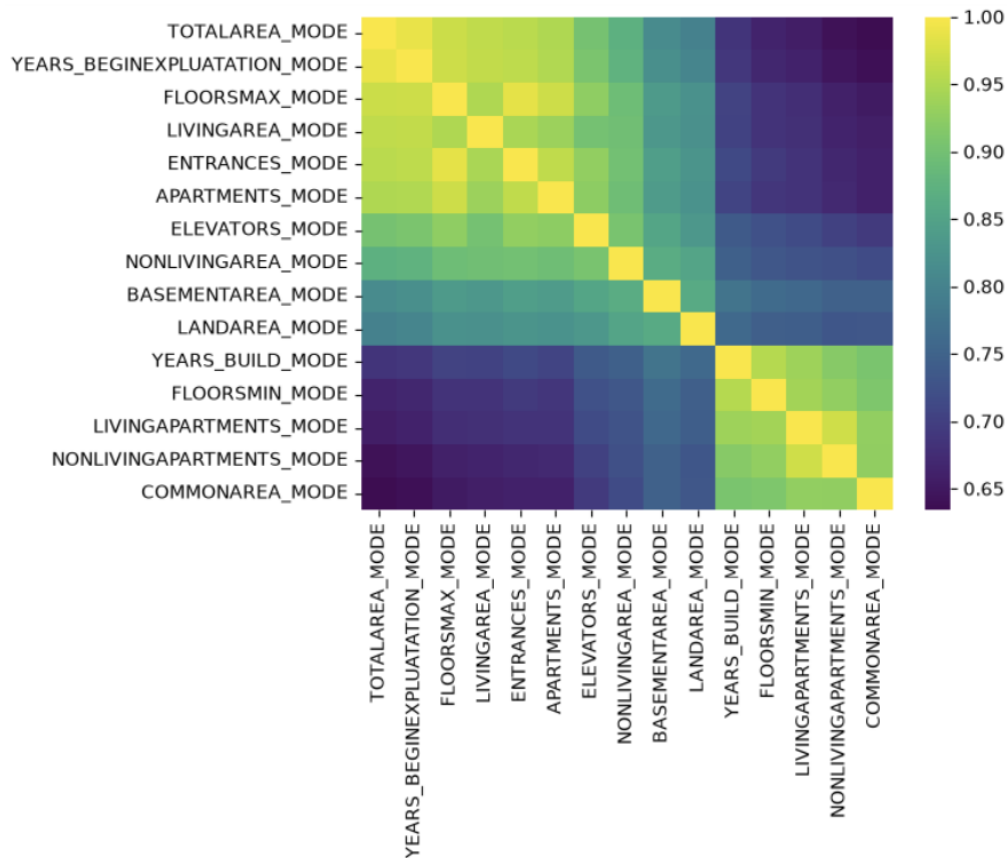


Figure 3. The missing value counts for many of the numerical columns are closely correlated, justifying the use of a single feature to record missingness. For each numerical variable, the missingness patterns of its mode, mean, and median statistics are perfectly correlated, so only the mode is shown.

5.2.4 Classifier hyperparameter tuning and training protocol

XGBoost is used as the primary classifier because it performs strongly on non-linear tabular data and naturally accommodates mixed numerical and categorical feature types. Gaussian naïve Bayes is included as a secondary model to provide a simple baseline and to validate that the observed patterns are not specific to a single classifier family.

Numerical features require imputation of missing values. Median imputation is applied because it is robust to outliers and maintains distributional stability across scenarios. XGBoost can ingest categorical features directly without encoding, and missing categorical values are replaced with a dedicated “missing” category, allowing the model to treat them as an additional level without distorting the underlying distribution. Binary features containing missing values are treated similarly by converting them to categorical form, ensuring that missingness is explicitly represented rather than implicitly encoded.

Gaussian naïve Bayes requires all numerical features to be on comparable scales, so numerical values are standardised using mean and standard deviation. Categorical features are one-hot encoded to ensure compatibility with the model’s independence assumptions.

XGBoost has a large hyperparameter space, and prior work (e.g., Song, 2025) indicates that a subset of parameters has substantial influence on performance. Grid search is used

to select hyperparameters, with implementation details provided in Appendix E, Notebook 07 (XGBoost Hyperparameter Tuning). To make tuning tractable across the experimental grid, hyperparameters are divided into two groups. The first group contains parameters expected to remain relatively stable across dataset sizes and imbalance ratios (`min_child_weight`, `max_depth`, `gamma`, `scale_pos_weight`). These are tuned first using grid search performed on a range of scenarios while holding the second group at default values. The second group contains parameters more sensitive to scenario characteristics (`colsample_bytree`, `learning_rate`, `reg_alpha`, `reg_lambda`, `subsample`). These are tuned subsequently using the optimised values from the first group. Both tuning stages use cross-validation to ensure stable parameter selection, with PR-AUC and ROC-AUC computed for each parameter selection. The number of boosting rounds (`n_estimators`) is selected based on sample size, with larger samples supporting more estimators. Increasing the number of trees improves model capacity but excessive boosting on small samples risks overfitting. Finally, the selected parameters for all scenarios are collated into a single file and examined for consistency and outliers. Final hyperparameters are provided in (Table 16). Gaussian naïve Bayes has no tuneable hyperparameters.

Training and evaluation are conducted in two stages, with implementation details provided in Appendix E, Notebook 08 (Training and Evaluation). First, each scenario-augmentation level is assessed using cross-validation (CV) on the training set only. This allows classification thresholds to be estimated without leaking information from the test set and provides an indication of the sensitivity of model performance to sample variation. Second, the selected models are evaluated on the holdout test set to assess generalisation and identify potential overfitting. CV is described below, and test set evaluation in the following section.

Unaugmented XGBoost training uses only real, in-scenario data. Stratified k-fold cross-validation (CV) is employed to preserve class balance within folds and to improve the stability of results, particularly in small-size tiers. Five folds are used to balance runtime with the informativeness of validation results. Synthetic samples used for scenario augmentation are next added to every training fold, using an identical synthetic dataset across folds. This design ensures reproducibility across repeated runs and prevents synthetic sampling variability from influencing model behaviour. Validation folds contain only real data so that augmentation affects training exclusively and does not inflate validation metrics. Early stopping is used to determine the best number of iterations.

Within each CV fold, the classification threshold is selected by choosing the probability cutoff that maximises the F1-score on the validation fold. F1-score is used as the optimisation target because it balances precision and recall, making it sensitive to minority-class performance. Imbalance-induced degradation primarily affects the minority class, while the majority-class typically remains stable due to its dominance in the confusion matrix. Because validation uses only real data, threshold selection is based entirely on real observations and remains unaffected by synthetic augmentation. Evaluation metrics include log loss, PR AUC, and ROC AUC, which are threshold-independent, and precision, recall, and F1-score, which depend on the tuned threshold. Validation-fold predicted probabilities are retained and used to calculate confusion matrices for selected scenarios, providing additional threshold-dependent insight.

Because CV tunes the classifier threshold within each fold, and because augmentation must be applied only to the training fold and not the validation fold, standard CV utilities cannot be used. The CV procedure is therefore implemented by manually iterating over the folds and calling dedicated training and evaluation functions.

5.2.5 Test set evaluation

Following cross-validation, all models are assessed against the same holdout test set, which introduces an additional computational requirement. The test set reflects the IR of the original data population, whereas each scenario simulates a different experimental IR. Because each scenario represents an underlying population with its own IR, test-set samples are weighted according to the ratios of the scenario class prevalence to the test-set class prevalence defined as:

$$Sample\ weight_{positive} = \frac{Positive\ class\ prevalence_{scenario}}{Positive\ class\ prevalence_{test}}$$

$$Sample\ weight_{negative} = \frac{Negative\ class\ prevalence_{scenario}}{Negative\ class\ prevalence_{test}}.$$

Weighting ensures that test-set evaluation reflects the scenario's simulated population IR, allowing direct comparison across scenarios despite differing class distributions. It is not required during cross-validation because the IR of validation folds already matches the training fold, and is only relevant for threshold-sensitive metrics: F1-score, precision, and recall.

For test-set evaluation, models are retrained using the full scenario real data combined with the synthetic data. Hyperparameters are identical to those used during cross-validation, except that early stopping is replaced by fixing the number of boosting rounds to the best iteration identified during cross-validation. As with cross-validation, the selected `scale_pos_weight` value is adjusted to reflect the augmented IR. To allow direct comparison with cross-validation results, the same evaluation metrics are calculated: log loss, PR-AUC, ROC-AUC, precision, recall, and F1-score. The classification threshold used is the median of the best thresholds found during cross-validation, which is a robust estimate. Probability predictions are again saved to enable confusion matrices to be calculated for selected scenarios.

5.2.6 Balanced training set performance across IRs

To isolate the impact of imbalance in the operating environment, one XGBoost classifier is trained using a balanced training set and then evaluated under simulated imbalance conditions. This allows the IR to be varied freely, and IR values from 0.001 to 1000 are used to examine the effect of both under and over-representation of the positive class. This experiment is equivalent to the earlier evaluations of imbalanced scenarios, except that data augmentation uses real data rather than synthetic and achieves a balanced distribution (IR 1).

The simulation is implemented by reusing the cross-validation probability predictions for the validation folds, and the scenario-level predictions for the test set and then recalculating threshold sensitive-metrics (F1-score, precision, and recall) using sample-weighted probabilities. Threshold tuning is again performed on the cross-validation fold

results, with the median threshold applied to the test set evaluation. Unlike with the scenario cross-validation, the balanced validation fold does not match the IR condition, so sample-weights must be applied. Fold sample-weights are calculated as:

$$\text{Fold sample weight}_{\text{positive}} = \frac{\text{Positive class prevalence}_{\text{IR condition}}}{0.5}$$

$$\text{Fold sample weight}_{\text{negative}} = \frac{\text{Negative class prevalence}_{\text{IR condition}}}{0.5}.$$

The denominator 0.5 reflects the IR 1 of the balanced training set. Similarly, for evaluation against the test set sample weights are calculated as:

$$\text{Test sample weight}_{\text{positive}} = \frac{\text{Positive class prevalence}_{\text{IR condition}}}{\text{Positive class prevalence}_{\text{test}}}$$

$$\text{Test sample weight}_{\text{negative}} = \frac{\text{Negative class prevalence}_{\text{IR condition}}}{\text{Negative class prevalence}_{\text{test}}}.$$

The implementation is fully detailed in Appendix E, Notebook 09 (Reports).

5.3 CTGAN-based data augmentation

This section addresses Objective 2 by analysing how CTGAN-based augmentation behaves under varying levels of data scarcity and class imbalance, and by determining whether an optimal augmentation level exists.

Three established approaches for generating synthetic data are: GANs; variational auto encoders (VAEs); and denoising diffusion probabilistic models (DDPM). All can model complex data distributions and are viable candidates for augmentation research. However, this study focuses on GAN-based augmentation because GANs have been most widely used in prior work addressing class imbalance. Since the research questions in this study specifically concern evaluating GAN-based augmentation under varying dataset size and imbalance conditions, GANs are selected as the generative architecture of interest.

The dataset used in this study is tabular and contains a heterogeneous mixture of categorical and numerical features. CTGAN is specifically designed for tabular data and incorporates mechanisms that make it suitable for this type of dataset, including explicit handling of categorical variables and modelling of mixed data types. CTGAN has been widely adopted in tabular synthetic-data research and is supported by a mature, well-documented Python implementation that enables reliable and reproducible experimentation. For these reasons, CTGAN is selected as the GAN architecture for generating synthetic minority samples in this study.

5.3.1 Data preprocessing

CTGAN is chosen as the GAN model because it is able to model datasets containing a combination of numeric and categorical features, which is required for the Home Credit dataset (Morelia, 2018) used for this study. CTGAN's architecture requires the datatype for every feature to be specified before training. Prior to training, the data must therefore

be represented using datatypes compatible with CTGAN's architecture: categorical, numerical (integer or float), or Boolean.

Binary-value features are treated as Boolean, regardless of whether they are originally stored as integers or strings. All other string-valued features are interpreted as categorical. Certain low-cardinality integer features that represent non-continuous, non-ordinal values are also treated as categorical, including:

- REGION_RATING_CLIENT
- REGION_RATING_CLIENT_W_CITY

All remaining integer features are treated as numerical but specified to be represented as integers, and float-valued features are treated as numeric.

5.3.2 Evaluation of synthetic data quality

Synthetic data fidelity is evaluated using a set of diagnostics appropriate for tabular GANs, as implemented in Appendix E, Notebook 05 (CTGAN Evaluation). For numerical features, correlations of mean, standard deviation, skewness, and kurtosis for each column between real and synthetic data are calculated. Each of these assesses in a single metric the alignment between real and synthetic data distribution statistics across all features. Kolmogorov-Smirnov statistic is calculated to provide feature-level analysis of distribution similarities, and pair-wise correlation is calculated to assess whether the patterns between features are maintained. For Boolean features, the difference in the probability of True is calculated, and any columns exhibiting mode collapse are listed. The latter is important because mode collapse is a known failure mode for GANs. For categorical features, the Jensen-Shannon divergence, Chi-Square, and Chi-Square Norm are calculated to determine the similarity of individual feature distributions, and the number of categories missing from the synthetic data are counted to detect possible mode collapse.

CTGAN utility is evaluated using train-real-test-real versus train-synthetic-test-real (TRTR-TSTR). Utility is ultimately the most important quality for synthetic data in a classification context because it helps determine whether models will benefit from augmentation. TRTR-TSTR trains two XGBoost classifiers, one using fully real data and the other with the positive class replaced by synthetic data. XGBoost is chosen because it is used for scenario evaluation and therefore provides the most relevant indicator of utility. Both datasets use an 80-20 stratified split of training and validation data and calculate metrics including log loss, PR-AUC, ROC-AUC, F1-score, precision, and recall, as these are the same metrics used in the experiment. Threshold tuning is also performed to calculate the tuned F1-score, precision, and recall. The results for both models are compared, with greater similarity in results indicating greater utility.

5.3.3 CTGAN hyperparameter tuning

Hyperparameter tuning for CTGAN is conducted with the goal of identifying a configuration that produces stable and representative synthetic samples across the range of minority-class sizes examined in the experimental grid, and not to exhaustively optimise generative performance. Four scenarios are selected to span this range as shown in Table 2. For each scenario, multiple candidate hyperparameter sets are

evaluated using the methods in the previous section, and a configuration is selected when it demonstrates good alignment between real and synthetic numerical feature statistics and shows no or minimal mode collapse in categorical and Boolean features. Hyperparameter tuning is fully detailed in Appendix E, Notebook 05 (CTGAN Evaluation) and selected hyperparameters are listed in Appendix A, Table 10.

Table 2. Four scenarios are selected for hyperparameters training to provide minority sample sizes spanning the experimental range. Note that CTGAN is trained only on minority samples from each scenario.

Size Tier	Imbalance Ratio	Minority Samples
Medium	81	545
Small	9	2,234
Small	3	5,586
Medium	1	22,234

Since the study examines the influence of dataset size and imbalance ratio rather than GAN optimisation, a single stable hyperparameter set was applied across all scenarios. This also ensures that differences in augmentation effectiveness arise solely from the experimental conditions rather than from variations in the generative model configuration.

5.3.4 CTGAN training and synthetic sample generation

For each imbalanced scenario a CTGAN model is trained on the minority samples and saved, as implemented in Appendix E, Notebook 04 (CTGAN Data synthesis). The scenarios are then augmented with synthetic minority samples generated by the models. For each augmentation level, a separate synthetic dataset containing the required number of samples is generated independently rather than cumulatively expanding a single synthetic pool.

Generating independent synthetic datasets ensures that each augmentation level represents a clean and isolated draw from the learned minority distribution. This avoids propagating any sampling anomalies, such as mode collapse or distribution drift, across augmentation levels. It also maintains the independence of experimental conditions, allowing differences in classifier performance to be attributed solely to the augmentation level rather than to patterns introduced by reusing synthetic samples. CTGAN does not allow a random state to be specified, and saved models regenerate the same samples each time they are opened. Therefore, to ensure augmentation levels use different samples, synthetic data for all required levels must be generated in one operation e.g. 50%, 100%, 200%, and 400%.

CTGAN can drop columns from exceedingly small synthetic datasets if they contain a single unique value. To structurally align the synthetic data with the real data these columns are reinstated using a random, constant value for categorical features or zero for numerical features. This could create noise, but these columns are uninformative and therefore a more sophisticated approach is not warranted.

5.3.5 Real-data augmentation control experiment

To provide an additional comparative benchmark, one size tier – small – is selected for a real-data augmentation experiment. In this analysis, the synthetic samples used to augment the scenarios are replaced with additional real observations drawn from the training pool. By comparing synthetic-data augmentation with real-data augmentation across varying IRs and augmentation levels, any underperformance can be attributed directly to the “syntheticness” of the added samples.

Evaluation of real-data augmentation scenarios follows the same procedure as synthetic-data augmentation, ensuring that differences in performance arise solely from the nature of the augmented data rather than from changes in methodology.

The real-data augmentation datasets are created in Appendix E, Notebook 02 (Experimental Scenarios). To avoid duplication, samples already present in the scenario are removed from the training pool. Datasets are then randomly sampled at the required augmentation levels.

5.3.6 Imputing missing values and one-hot encoding

During training and evaluation, missing values for numerical data are imputed for real data. The imputer is trained on the real training data only to avoid information leakage, then applied to the validation fold or test set. By contrast, the synthetic data applied to each scenario is generated from a model that learns from the same training set which first has missing values imputed and therefore contains no missing values. This approach is valid because the imputer applied to the training set prior to CTGAN training is based exclusively on the scenario data.

Gaussian naïve Bayes requires categorical features to be one-hot encoded (OHE). As with the imputation function, the encoder is trained on the training set and only then applied to real and synthetic training data, the validation fold or the real test data. The synthetic training data is guaranteed to include only categories present in the training data because the model which generated it used the same training set. However, the validation fold and test set may include additional values, which the encoder converts to zero-vectors.

5.4 Balanced-world F1 recovery

This section addresses Objective 3 by evaluating the extent to which CTGAN-generated samples restore classifier performance relative to the balanced counterfactual, using the normalised recovery metric.

To quantify degradation in performance experienced in simulated scenarios, we define the metric Balanced-world F1 performance loss as:

$$F1\ loss = F1\ performance_{balanced} - F1\ performance_{0\% \ aug}.$$

Performance recovery achieved relative to the un-augmented baseline is given by:

$$F1\ recovery = F1\ performance_{aug} - F1\ performance_{0\% \ aug}.$$

Normalised recovery, defined as the degradation recovered through augmentation as a proportion of performance lost, is given by:

$$F1\ recovery_{norm}\% = \frac{F1\ recovery}{F1\ loss} \times 100\%.$$

Balanced-world F1 recovery quantifies the effectiveness of augmentation relative to the performance that was lost when imbalance was introduced into the population. F1 is threshold-dependent, so Balanced-world F1 loss reflects both the class separability inherent in the data and the distribution of the test set. This makes it an appropriate metric for quantifying degradation and recovery in real-world imbalanced evaluation.

F1 recovery is implemented in Appendix E, Notebook 08 (Training and Evaluation), and is calculated as a final step after all scenarios have been executed because it requires results from the balanced scenario, the unaugmented imbalanced scenario, and the augmented scenario.

6.0 Results

6.1 Degradation under imbalance and data scarcity (RQ1)

This section addresses Objective 1 by examining classifier behaviour under controlled class imbalance and varying sample size.

6.1.1 Sample weighting of test set

Because a common imbalanced test set with a fixed IR was used to evaluate all scenarios, samples were weighted according to the ratio of the scenario to the test set. This was not required for cross-validation where validation folds retained the IR of the scenario. Figure 4 shows that Evaluation results against the test set were consistent with cross-validation only after sample weightings were applied, validating this approach.

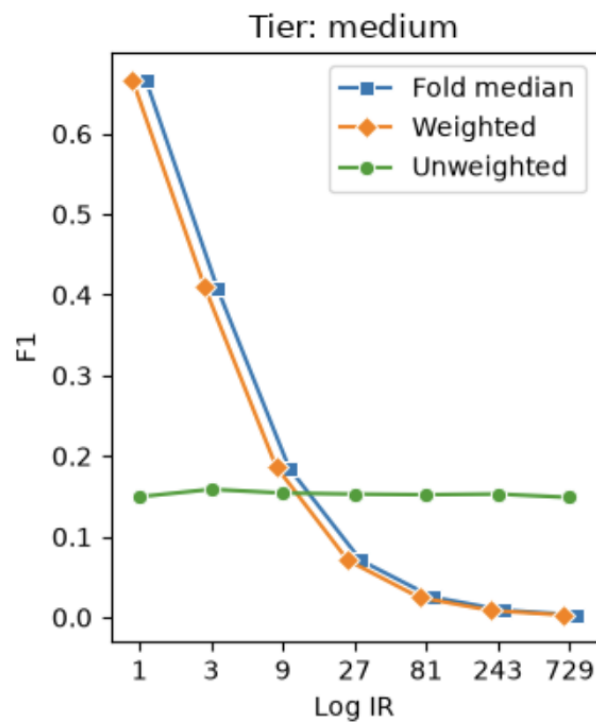


Figure 4. Classifier performance evaluated using cross-validation, weighted test set and unweighted test set. Weighted test set evaluation was consistent with cross-validation.

6.1.2 Scenario results

Figure 5 shows confusion matrices for the un-augmented small tier across increasing IR values. At IR = 1 (balanced), the classifier predicts the positive class more frequently than the negative class, and although this produces a noticeable number of false positives, positive precision remains reasonable (0.627). As imbalance increases, the distribution of predictions shifts. By IR 9, both positive precision and recall have fallen below 0.5, while the negative class becomes increasingly dominant in the predictions. From IR 81 onwards, positive predictions collapse almost entirely, and the confusion matrices become dominated by true negatives and false positives, indicating that under high imbalance the classifier effectively stops correctly identifying positive cases.

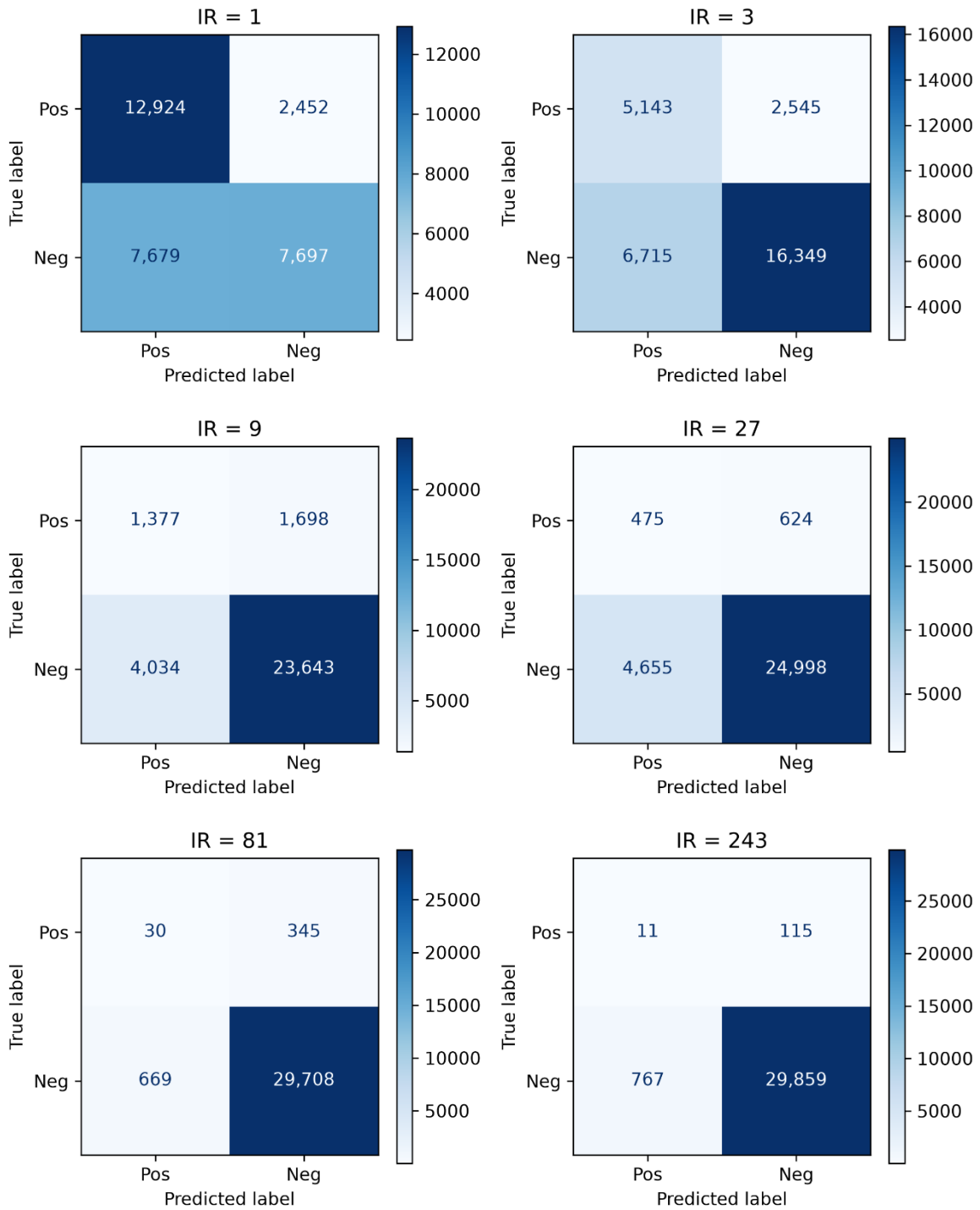


Figure 5 Beyond IR 81 the classifier fails almost entirely to predict the positive class.

Figure 6A quantifies the effect of imbalance, showing how F1 varies with the log of IR across all size tiers. F1-score consistently falls with increasing IR, with only a small gap between tiers when balanced, a wider gap at moderate IR levels, and convergence again towards zero at extreme IRs as all tiers approach F1 = 0.

Although the effect of sample size is weaker than IR and is not easily visible in confusion matrices, it becomes clear when F1 is plotted against the log of sample size (Figure 6B).

This shows that, for all IR levels, F1 increases approximately linearly with $\log(\text{sample size})$, with slopes between 0.06 and 0.12 per 10-fold increase in training size (Table 3).

Together, Figure 6A and Figure 6B show that imbalance exerts the dominant influence on classifier behaviour, while sample size provides a secondary but consistent performance gain across all IR levels. They also highlight that while additional data improves performance, it cannot compensate for extreme imbalance.

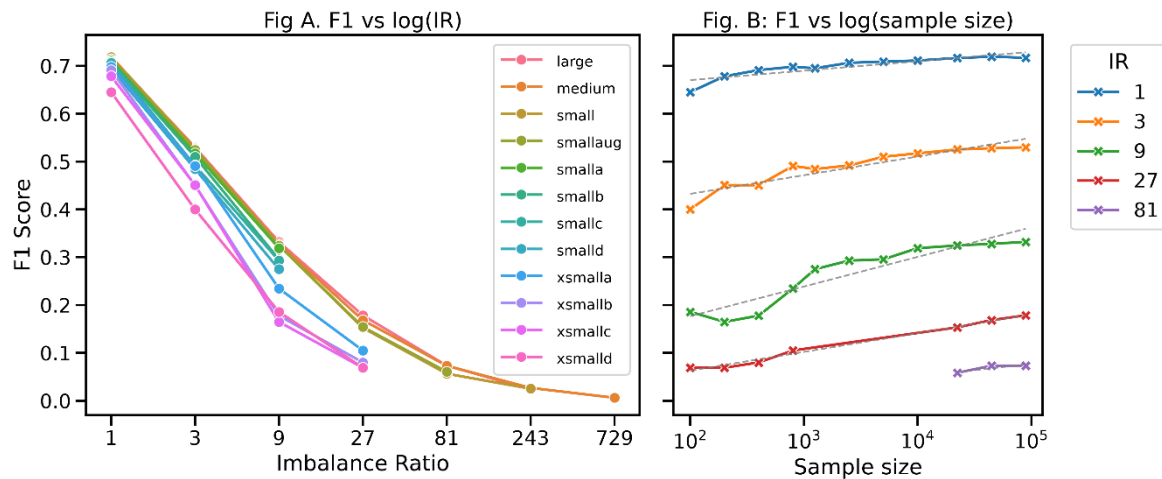


Figure 6A. F1 drops with increasing IR. **Figure 6B.** F1 increases with increasing samples size, but effect is weaker than for IR.

Table 3. Gradient of F1 versus $\log(\text{sample size})$

Imbalance Ratio	Gradient
1	0.020
3	0.039
9	0.062
27	0.039
81	0.027

While Figure 6 illustrates the overall trends, Table 4 summarises the absolute F1 loss (balanced minus imbalanced) across imbalance ratios and sample-size tiers. This provides a direct numerical measure of degradation.

Table 4. F1 performance loss compared to the counterfactual balanced scenario increases sharply with IR and gradually with decreasing sample size.

Imbalance Ratio	Sample size			
	Large (89,368)	Medium (44,684)	Small (22,342)	X-Small (800)
3	0.187	0.191	0.191	0.207
9	0.385	0.391	0.392	0.464
27	0.538	0.551	0.563	0.593
81	0.644	0.646	0.658	-

Imbalance Ratio	Sample size			
	Large (89,368)	Medium (44,684)	Small (22,342)	X-Small (800)
243	0.693	0.692	0.693	-
729	-	0.713	-	-

Other metrics also vary systematically with imbalance. Recall, precision, and the best threshold decrease approximately linearly with $\log(\text{IR})$, while ROC-AUC and PR-AUC show similar near-linear trends. Log-loss follows an approximately linear pattern up to IR 27, after which it becomes more variable. Full plots for all metrics are provided in Appendix E (Figure 11).

6.1.3 Balanced training model evaluated in imbalanced worlds

A model trained on a perfectly balanced dataset was evaluated across a sweep of increasingly imbalanced test worlds, and the results are shown in Figure 7. Despite the training distribution being balanced, classifier performance still degrades as the imbalance ratio (IR) of the evaluation world increases. Once the test population becomes dominated by the negative class ($\text{IR} \gg 1$), F1-score collapses, reaching near-zero at IR 1000. Conversely, when the evaluation world contains an abundance of positive samples ($\text{IR} < 1$), the model performs extremely well: F1-score increases steadily as IR decreases and reaches 0.995 at IR 0.01.

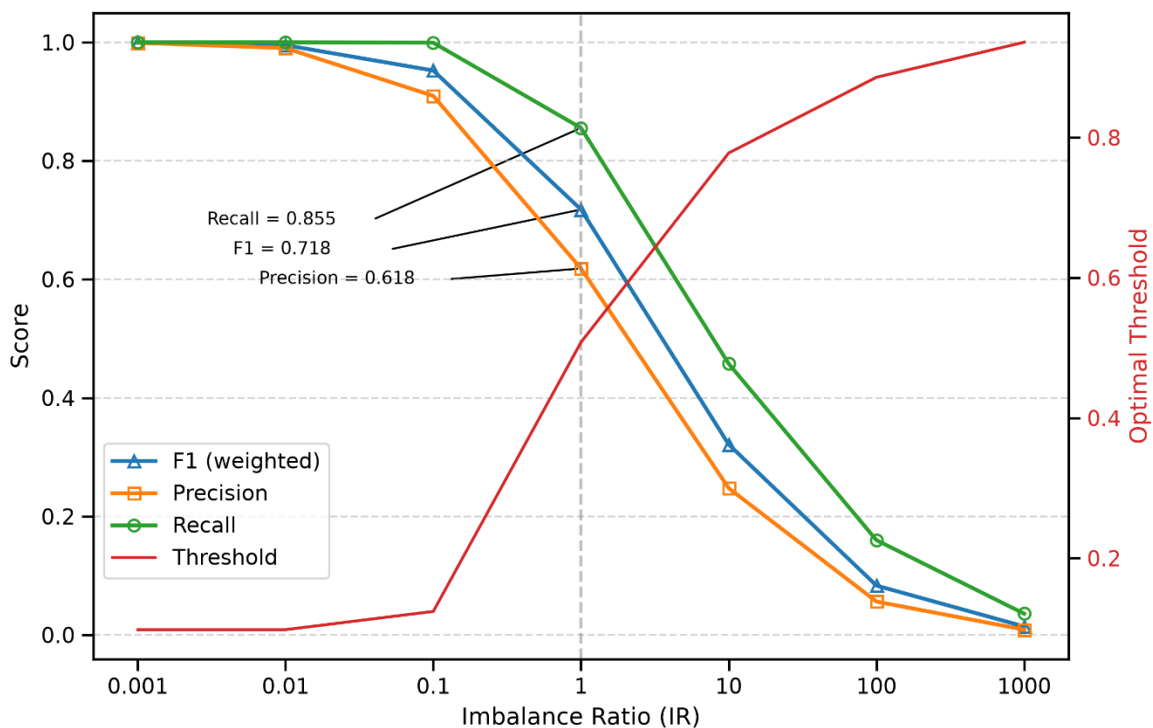


Figure 7. A model trained on a balanced dataset performs well when evaluated in worlds with abundant positive samples ($\text{IR} < 1$), but its performance deteriorates as the evaluation world

becomes increasingly dominated by negative samples. Balancing the training set does not prevent performance collapse under extreme evaluation imbalance.

6.2 Effect of augmentation (RQ2)

6.2.1 CTGAN fidelity evaluation

Numerical fidelity was assessed using column-wise distribution comparison, the Kolmogorov–Smirnov statistic, and pairwise correlation. Column-wise distribution similarity was evaluated by calculating the Pearson correlation between distribution statistics (mean, standard deviation, skewness, and kurtosis) for each column in the real and synthetic data. Results are shown in Table 5. Kolmogorov–Smirnov statistic was evaluated by aggregating values across the columns, with results summarised Table 6 and listed in full in Appendix A (Table 11). Pairwise correlation fidelity was evaluated by computing the difference between real and synthetic pairwise correlation coefficients, with the aggregated results shown in Table 7 and the correlation heatmaps shown in Figure 8.

Table 5. Correlation between real and synthetic numerical features increases with increasing sample size until the largest size.

Sample size	Mean	Std	Skew	Kurtosis
545	0.997	0.997	0.605	0.256
2,234	0.997	0.995	0.738	0.543
5,586	0.999	0.996	0.813	0.672
22,342	0.998	0.672	0.176	-0.027

Table 6. Kolmogorov–Smirnov (KS) statistic mean and standard deviation quantifying the aggregated distance between real and synthetic numerical populations.

Sample size	Mean KS	Std KS	Max KS
545	0.274	0.160	0.543
2,234	0.295	0.186	0.673
5,586	0.293	0.188	0.653
22,342	0.308	0.192	0.628

Table 7. Difference between real and synthetic numerical pair-wise correlation coefficients.

Sample size	Mean ABS difference	Std ABS difference	Max ABS difference
545	0.178	0.229	1.065
2,234	0.149	0.215	1.024
5,586	0.093	0.140	0.983
22,342	0.084	0.118	0.929

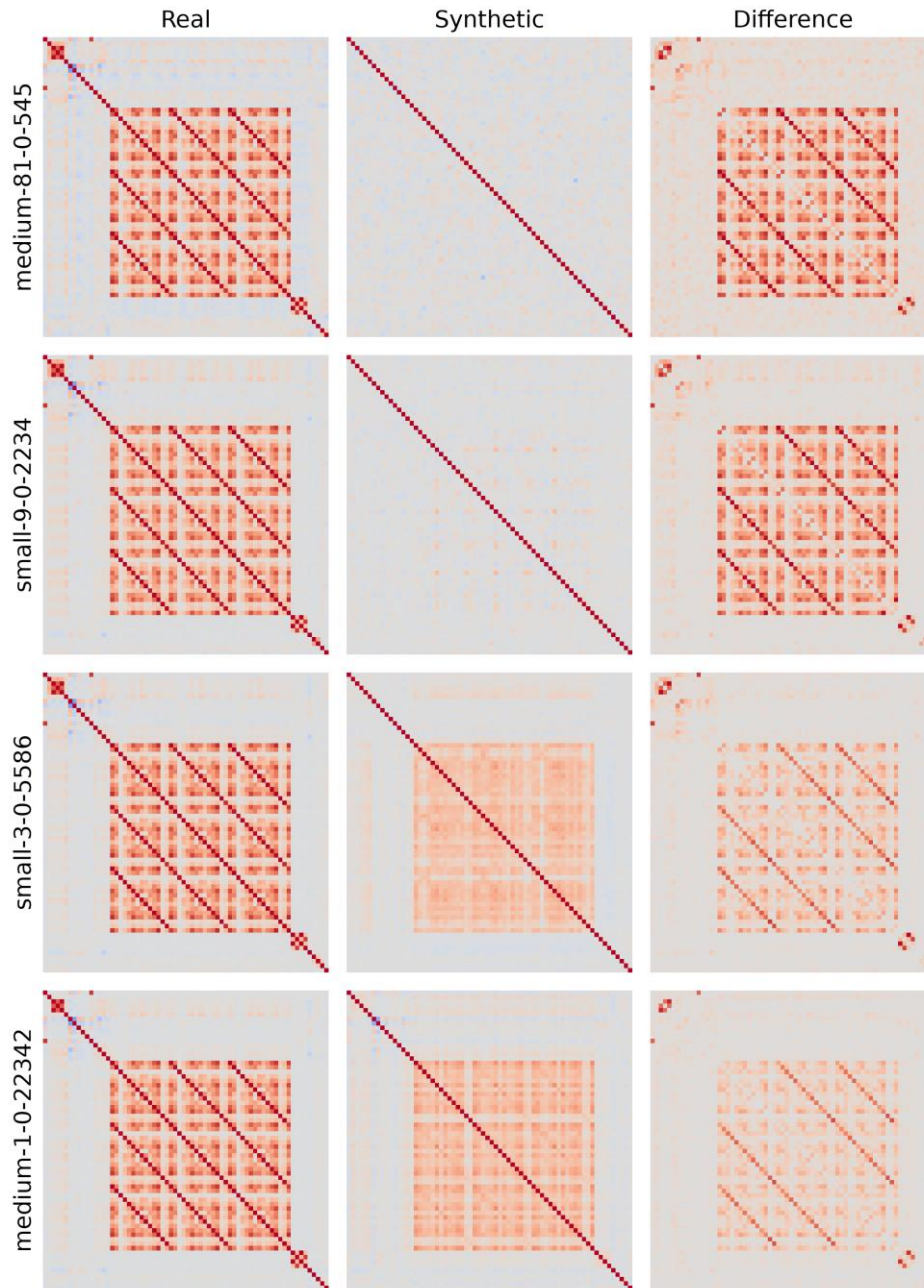


Figure 8. Correlation heatmaps show that below 5,586 samples, CTGAN is unable to replicate patterns between numerical features. These become apparent only from 5,586 and strengthen slightly at 22,342.

Boolean fidelity was evaluated by comparing the probability of True in the real and synthetic data and identifying instances of mode collapse. Differences ranged from 0.004 to 0.028 and no mode collapse was observed in any sample (Table 12).

Categorical fidelity was evaluated using Jensen–Shannon divergence (JSD), Chi-square, Chi-square norm, and the number of missing categories for each column. The range of JSD was relatively constant across sample sizes, while Chi-square and Chi-square norm increased with sample size (Table 8). Only one column – ORGANIZATION_TYPE, which contains over fifty categories — exhibited mode collapse, missing just over 10% of categories at smaller sample sizes. At 22,342 samples, all categories were generated.

Overall, categorical fidelity metrics showed consistent JSD values and increasing Chi-square measures with sample size, with mode collapse limited to a single high-cardinality column. Categorical results are listed in full in Appendix A (Table 13).

Table 8. Range of JSD is relatively constant while Chi-square and Chi-square norm vary across columns and samples sizes.

Sample size	Jenson-Shannon Divergence	Chi-Square Norm	Chi-Square
545	0.000 – 0.013	0.288 – 3.828	1.730 – 36.042
2,234	0.000 – 0.011	1.288 – 22.623	9.016 – 86.010
5,586	0.000 – 0.006	1.170 – 136.077	5.448 – 272.154
22,342	0.000 – 0.008	5.386 – 281.119	59.062 – 1686.711

Note. Table shows results excluding ORGANIZATION_TYPE column, which has over fifty categories and exhibited mode collapse in small sample sizes.

6.2.2 CTGAN utility evaluation

CTGAN utility was evaluated using train-real-test-real versus train-synthetic-test-real (TRTR-TSTR). Four scenarios were evaluated, each using a stratified 80-20 train-test split, with results for the largest scenario shown in Table 9. XGBoost with early stopping was used and metrics were calculated including log loss, PR-AUC, ROC-AUC, F1-score, precision, and recall. Threshold tuning was also performed to calculate tuned F1-score, precision, and recall. Results across a range of sample sizes and IRs (Table 15) showed that, while the real data always learned to classify samples to some degree, the synthetic data failed consistently. The best threshold for real data was between 23 and 179, but the synthetic never improved after the first iteration, except for the small IR3 scenario which peaked at 112 iterations. For all scenarios, the ROC-AUC remained close to 0.5 and after threshold tuning, the best threshold was 0 or 1, yielding recall of 1 and precision between 0.012 and 0.5. In contrast, the real data had tuned thresholds between 0.017 and 0.215, ROC-AUC of 0.693-0.751, recall of 0.165-0.861, and precision of 0.065-0.615.

Table 9. Results for the 22k balanced sample showing that synthetic data fails to learn negative-class patterns.

Metric	TRTR	TSTR
Number of estimators	1200	1200
Best iteration	179	1
PR-AUC	0.741	0.489
ROC-AUC	0.751	0.496
Log loss	0.637	0.802
Maximum probability	0.913	0.458
Precision (t=0.5)	0.771	0.000
Recall (t=0.5)	0.422	0.000
F1-score (t=0.5)	0.546	0.000

Metric	TRTR	TSTR
Best threshold	0.215	0.000
Precision (tuned)	0.616	0.500
Recall (tuned)	0.861	1.000
F1-score (tuned)	0.718	0.667

6.2.3 Effect of Synthetic-data augmentation

Synthetic-data augmentation was evaluated at levels up to 400% across scenarios of varying size and imbalance using XGBoost and Gaussian naïve Bayes, and results are shown in Figure 12. For XGBoost, augmentation consistently reduced log-loss across all scenarios by between 0.05 and 0.35, with the largest reductions occurring at higher augmentation levels (Figure 9). The tuned classification threshold also fell with increasing augmentation, typically by 0.1 to 0.3. Small increases in ROC-AUC and PR-AUC (0.005–0.015) were observed in a subset of conditions, including small IR 27 and IR 81, medium IR 3 and IR 81, and large IR 81 and IR 243. Other metrics and scenarios showed minimal improvement or slight degradation with augmentation. Overall, augmentation produced modest and inconsistent effects outside of log-loss and threshold, and no systematic pattern was evident across IR or sample-size tiers.

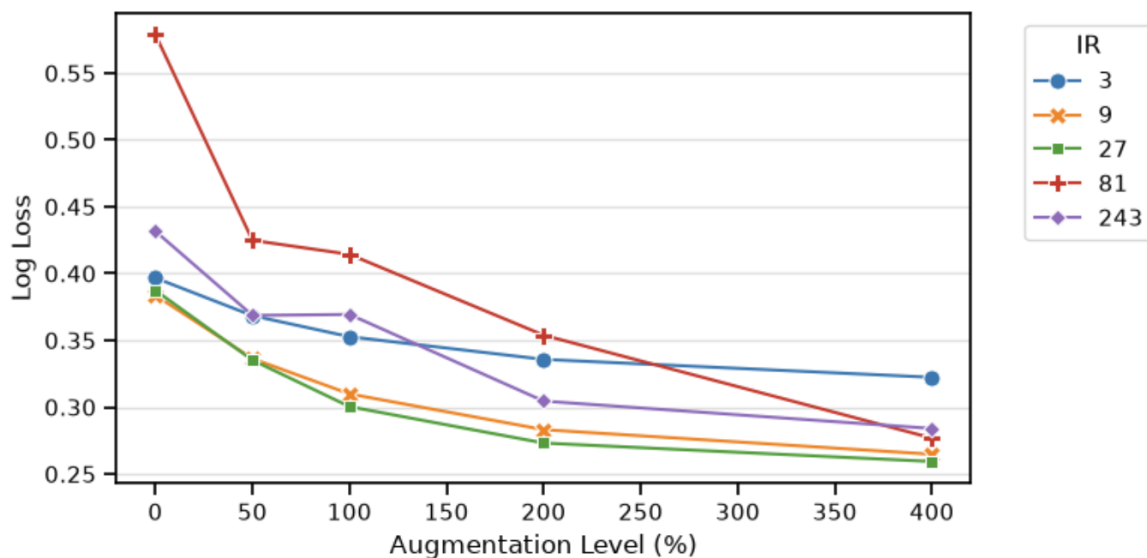


Figure 9. Small scenario shows log-loss falling with augmentation level, as exhibited by all scenarios.

Classification thresholds were unstable for augmented Gaussian naïve Bayes, varying irregularly across augmentation levels, but the remaining metrics behaved similarly to those of XGBoost, as shown in Figure 13.

6.2.4 Augmentation with real data

A supplementary experiment was conducted in the small tier in which augmentation was performed using additional real observations rather than synthetic samples. This served as a control to determine whether the lack of improvement under synthetic

augmentation could be attributed to the CTGAN synthetic generation process. As with synthetic-augmentation results, real-data augmentation did not improve performance: although log-loss and classification threshold reduced with increasing augmentation, the remaining metrics behaved similarly to the synthetic-augmentation case and showed no meaningful gains. Full results are shown in (Figure 14).

6.3 F1 Recovery (RQ3)

F1-recovery was also evaluated across augmentation levels. Because augmentation did not improve performance in any condition, F1-recovery showed no systematic behaviour and did not provide additional insight beyond the primary metrics. Visualisations for XGBoost (Figure 15) and Guassian naïve Bayes (Figure 16) are included in appendices.

Insufficient minority samples were available to construct the large balanced scenario required to compute F1-recovery. To allow the metric to be defined, the balanced-scenario F1 values for both classifiers were estimated from the medium and small scenarios, which were nearly identical. Because sample size had only a modest effect on performance for all IRs, using estimated value did not impact the experimental outcomes or interpretation of recovery.

7.0 Discussion

This chapter examines the findings presented in the results chapter and discusses how they address the project aims and research questions.

7.1 Degradation under imbalance and scarcity

The aim of RQ1 was to characterise the effects of imbalance and data scarcity on classifier performance and determine the amount of degradation experienced as these factors increase relative to a counterfactual balanced scenario.

Confusion matrices for each size tier (e.g. small tier in Figure 5) show that as IR increases, the classifier predicts the positive class less often and less accurately, with recall falling to 0.1 at IR 81 and precision to 0.1 at IR 27. Figure 6A shows that F1 falls approximately linearly with $\log(\text{IR})$ at moderate IRs, then more gradually as F1 approaches 0 at extreme IR levels. F1 performance loss in Table 4 quantifies the F1 performance loss experienced by each scenario compared to the balanced scenario. Consistent with Figure 6A, it shows that the rate of F1 loss is highest close to IR 1 then falls as IR increases. It also shows that F1 loss is similar across size tiers, converging at extreme levels – IR 243. These results are predicted by several studies (He and Garcia, 2009; Fernandez et al., 2018).

The effect of imbalance in the operational population was also analysed by evaluating models trained on a balanced training set (Figure 7). Performance declined in the same way as for models trained on imbalanced data, with F1 decreasing roughly linearly with $\log(\text{IR})$ at moderate IR and more gradually at extreme levels. The similarity is shown clearly in Figure 10 where results from imbalanced training are overlaid with balanced training. Together with the imbalanced training results, this suggests that the prevalence of majority samples in the operational population is the major determinant of classifier

performance when measured using F1-score. Had performance degradation been solely due to majority class bias in the training set, the balanced training set would have recovered some of the loss, but this was not the case.

It is also possible that the classifier simply failed to learn a meaningful decision boundary. However, this appears unlikely because in the balanced scenario it achieved an F1-score of approximately 0.7, and while only moderate, this remains substantially above the 0.5 baseline of an uninformed classifier. This indicates that the model was able to extract genuine structure from the data. The decision to use a single missingness indicator for many numerical features, chosen to avoid a large increase in dimensionality, may have hidden useful information and adversely affected performance. However, this would not be expected to alter the observed patterns of degradation. Similarly, utilising feature engineering and including the Home Credit Risk supplementary tables would have improved performance but not contributed to the objectives of the study.

Data scarcity was also found to degrade classifier performance, but to a lesser degree than IR (Figure 6B). Table 3 shows that increasing sample size by a factor of ten raised F1 by only 0.02-0.06, demonstrating that increasing sample size does not easily overcome the impacts of imbalance.

The curves in Figure 6 further show that the effects of imbalance and data scarcity are additive rather than multiplicative: F1-log(sample size) curves were shallow and exhibited no discernible patterns across IR levels, and F1-log(IR) curves had similar shapes across all sample-size tiers. Fernandez et al. (2018) also explain how data scarcity exacerbates imbalance-induced learning difficulties by further underrepresenting clusters within the minority class.

The weighted test-set evaluation aligned closely with the cross-validation results (Figure 4), indicating that weighting successfully corrected for the mismatch between the test set IR and the scenario IR. This consistency confirms that the observed patterns reflect genuine scenario behaviour rather than artefacts of the evaluation procedure and supports the validity of the weighting approach.

7.2 Optimum augmentation level

The aim of RQ2 was to evaluate whether synthetic augmentation could improve classifier performance and whether an optimal augmentation level exists across different data conditions.

Across four training sizes (545 to 22,342) CTGAN produced synthetic data with moderate fidelity. Numerical features showed strong correlations between real and synthetic means and standard deviations for the smaller tiers, with correlations weakening at the largest tier, suggesting that CTGAN may struggle to maintain diversity as sample size increases. KS statistics indicated moderate distributional fidelity (mean KS 0.27–0.31), with higher values for skewed or long-tailed variables. KS statistic increased slightly with sample size but this likely reflects increasing measurement precision rather than a change in divergence. Correlation heatmaps (Figure 8) showed that CTGAN struggled to replicate distributions for sample sizes below 5,586, and that the correlations improved only slightly from 5,586 to 22,342. One explanation is that deep-learning based generative model like CTGAN require a minimum data volume to train networks. However, it is also

possible that fixing the hyperparameter selection across the sample size range optimised performance for the middle of the range. For example, the chosen network architectures of 128-128 for the generator and discriminator may have been too large for the small samples – resulting in overfitting – and too small for large sample – resulting in high bias.

Boolean features exhibited good fidelity, with close alignment in class probabilities and no evidence of mode collapse. Categorical fidelity was also strong: Jensen–Shannon divergence values were extremely low (0.000–0.011), indicating close alignment between real and synthetic distributions. Chi-square norms increased with sample size, which is typical, and did not indicate poor fidelity. Only one feature (ORGANIZATION_TYPE) showed partial collapse at small sample sizes, likely due to its very large number of categories (41–58), which makes full coverage difficult in small datasets.

While fidelity results were reasonable overall, utility was poor. TRTR versus TSTR experiments showed that models trained on synthetic data could not distinguish real positives from synthetic negatives, and collapsed to naive behaviour, predicting the positive class for almost all samples. In contrast, models trained on real data displayed a discriminatory ability. Xu et al. (2019) also observe that even high-fidelity synthetic samples may have poor utility for downstream learning tasks. One possible explanation is that, although CTGAN reproduced feature-level distributions, it did not capture the minority-class structure required for effective classification. Kotelnikov et al. (2023) also show that GANs can struggle to capture complex tabular distributions, resulting in poor sample utility. Pair-wise correlations suggest that CTGAN learned numerical feature relationship reasonably well for the 5,586 and 22,342 sizes, but this does not confirm that class separability was preserved. Alternatively, it is possible that training CTGAN solely on minority samples prevented it from representing a realistic boundary. Although this approach avoided majority-class bias, it produced samples clustered around the densest minority regions and away from boundaries where samples are most informative. As a result, replacing the real minority samples with synthetic ones removed the discriminative structure from the dataset.

Augmentation did not produce meaningful improvements. As shown in Figure 9, log-loss decreased with increasing augmentation, but precision, recall, and F1 fluctuated only slightly and showed no consistent pattern. This might be attributable to the poor utility of synthetic minority samples; however, augmentation with real data produced similarly weak results (Figure 14). This suggests that, for this dataset, the challenge is not the imbalance in the training data but the imbalance in the population in which the model must operate. Additionally, the result contradicts the intuition applied to imbalanced-learning problems that simply balancing the training data will reliably improve classifier performance due to the elimination of negative class bias. This interpretation is supported by comparing balanced-training models evaluated under different imbalance conditions with models trained directly on imbalanced data (Figure 10). Performance in both cases was almost identical across IRs, except for small differences at IR 27 and 81 (<0.05). These results indicate that augmentation—synthetic or real—does not meaningfully alter classifier behaviour for this dataset. This is consistent with Zhao et al. (2025) who assert that classification is more difficult when class-defining features overlap and boundaries are unclear, and add that when classes overlap significantly, adding additional samples is unhelpful. He and Garcia (2009) also suggest that data complexity is the major determinant of impaired learning, and that imbalance simply amplifies this.

Instead, methods that directly improve boundary definition or increase minority-class diversity may be more effective than simple augmentation. For example, Zhao et al. state that selectively removing samples in areas of overlap can improve performance.

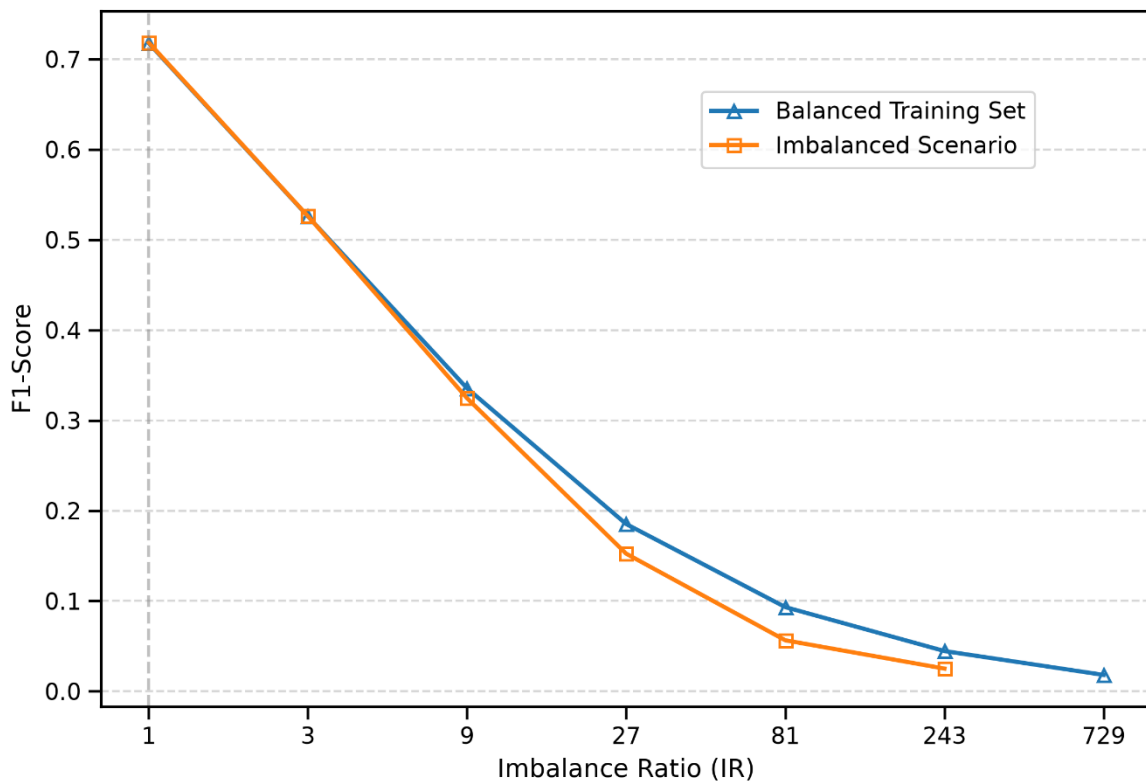


Figure 10. Models trained on a real, balanced training set and evaluated in imbalanced settings (increasing by factors of three to match the imbalanced scenarios) perform marginally better than imbalanced models at IR 27 and 81 but converge at higher and lower values.

Given that data augmentation for this dataset was found not to improve classifier performance, it was ultimately not possible to establish an optimum level.

7.3 Extent of performance recovery with augmentation

The final research question aimed to evaluate the extent to which imbalance-induced degradation in classifier performance could be restored using data augmentation, relative to a balanced counterfactual. The results showed that augmentation of the minority class using synthetic data generated by CTGAN, and even real data, did not meaningfully improve classifier performance. These results support Engelmann and Lessmann (2021) who find that GAN-based augmentation does not consistently improve classifier performance, and Zhao et al. (2023) who reported that while CTAB-GAN+ – which is more advanced than CTGAN – did improve performance, results were dataset-specific. Although the proposed F1 recovery metric was implemented and calculated for all scenarios, because it is based on recovered loss, it also did not reveal any informative patterns.

8.0 Conclusions

8.1 Overall conclusions

This project aimed to characterise how class imbalance and sample size affect classifier performance, and to evaluate the extent to which GAN-based data augmentation can restore performance under these conditions. The study achieved this aim by developing an experimental framework that independently varied imbalance ratio and sample size, enabling performance degradation to be quantified relative to a counterfactual balanced scenario. It fulfilled the first objective by demonstrating how performance declines as imbalance increases and sample size decreases. It fulfilled the second objective by analysing augmentation behaviour across all IR-sample-size combinations, showing that no optimal augmentation level emerged. It fulfilled the third objective by implementing a recovery metric and evaluating CTGAN-based augmentation across all scenarios, finding that recovery was effectively zero. Together, these outcomes confirm that all project objectives were met.

This study addressed three research questions. First, it quantified performance degradation relative to a balanced counterfactual and found that loss increases sharply with $\log(\text{IR})$ and more gradually with $\log(\text{sample size})$. Second, it examined whether an optimal augmentation level exists, and no such optimum was identified because augmentation produced no meaningful improvement in any scenario. Third, it assessed the extent to which CTGAN-based augmentation can compensate for imbalance-induced degradation, and recovery was effectively zero across all conditions.

The findings show that imbalance ratio is the dominant driver of classifier performance, while sample size has a smaller but predictable effect. These patterns were consistent across both XGBoost and Gaussian naïve Bayes, indicating that the underlying relationships are not model-specific. CTGAN generated samples with reasonable fidelity but low utility, and neither synthetic nor real-data augmentation improved classifier accuracy. By evaluating multiple augmentation levels across all combinations of imbalance ratio and sample size, the study demonstrated that increasing minority-class representation alone does not overcome the underlying class separability of the dataset.

The project contributes a generalisable experimental framework capable of jointly analysing the effects of IR, sample size, and augmentation level. Because the framework controls the fundamental parameters of tabular classification – imbalance ratio, sample size, and augmentation level – it is generalisable to any tabular dataset and classifier assessing the sensitivity of performance to imbalance and data scarcity. As imbalance and sample size are structural properties of the dataset rather than characteristics of the underlying distribution, the empirical findings regarding their effects on performance are also generalisable and expected to hold across other domains of tabular data. The framework also introduces the normalised F1 recovery metric, which allows augmentation benefit to be compared across scenarios with different imbalance levels. This metric should be generalisable to other datasets and models, although was not fully evaluated here because augmentation did not improve performance.

Together, these contributions provide practitioners with a structured way to evaluate whether augmentation or further data collection is likely to be most effective, subject to

the limitations outlined in the following section. The results also demonstrate that restoring balance in the training set does not improve performance when evaluation occurs under imbalanced conditions, highlighting that the operating distribution, not the training distribution, determines achievable performance.

Overall, the study clarifies the limits of GAN-based augmentation for imbalanced tabular data and establishes a basis for future research into methods that address class separability rather than minority-class quantity alone.

8.2 Limitations

The experimental framework requires a classifier to be trained, and hyperparameters tuned for every combination of IR and sample size. Depending on the model, this can be computationally costly, and finite time constrains the number of scenarios that can be evaluated. Although each augmentation level also requires a separate classifier, hyperparameters can be reused across augmentation levels. Augmented scenario training times can be reduced by warm-starting models pretrained on the unaugmented data, but only for classifiers that support incremental updates or fine tuning.

As with classifiers, a GAN must be trained for every scenario. Furthermore, pair-wise correlation results suggest that hyperparameters may require tuning for specific sample sizes, or at least a representative range. GAN training is significantly more expensive than classifier training (Table 14), so these requirements further limit the size of the experimental grid that can practically be explored.

CTGAN was trained only on minority class samples; in part because CTGAN does not natively permit conditional synthetic data generation, but primarily to avoid majority-class bias. This approach likely impaired the utility of the synthetic data by preventing the minority-majority class boundary from being represented effectively. Training a GAN on both minority and majority samples would generate minority samples that contain a realistic boundary structure and better support classifier learning.

While the F1 recovery metric provides a consistent way to compare augmentation effectiveness across scenarios, its interpretability is limited because manipulation of the training data cannot alter the evaluation environment and, therefore, can never fully overcome the degradation due to imbalance. Moreover, methods for addressing imbalance may improve performance by enhancing class separability rather than solely by reducing majority-class bias introduced by imbalance. This means that the metric combines benefits from both effects rather than purely measuring imbalanced-induced loss recovery.

Although the implementation allows multiple bootstraps, each scenario was executed only once due to computational constraints, limiting the ability to quantify run-to-run variance for GAN training and classifier performance. This does not affect the structural conclusions but restricts the precision with which variability can be assessed.

8.3 Future work

8.3.1 Broader datasets and model families

Further insights would be gained by applying the experimental framework to datasets from different domains and to alternative classifier families including deep learning models. Extending the scope of generative modelling beyond CTGAN – such as DDPM, TVAE or CTAB-GAN+ – would also broaden the evidence base. Exploring a wider range of datasets and models would strengthen generalisability of the findings and deepen understanding of augmentation behaviour in different contexts.

8.3.2 Majority and minority GAN training

CTGAN produced samples with low utility, due in part to being trained only on minority samples. Future studies should train generative models on both minority and majority samples so that the model learns the minority-majority class boundary. Synthetic minority samples could then be generated using conditional sampling, if supported by the model, or using random sampling and class-based filtering if not. Training on both classes also increases the data volume, making shared feature semantics easier for the generator to learn.

8.3.3 Informed sampling

The inability to improve classifier performance by simply correcting class imbalance suggests that informed sampling strategies, such as those used in Borderline-SMOTE (Fernández, 2018), could be adapted for generative augmentation. Rather than producing large volumes of synthetic data and augmenting indiscriminately, a generative model could be coupled with a sampling mechanism that targets minority–majority boundary regions. Conditional generators can produce such samples directly; non-conditional models could generate a large pool of synthetic points and then select those lying near the decision boundary. This approach preserves SMOTE’s boundary-focused augmentation while avoiding its known failure modes, such as generating unrealistic points or interpolating between unrelated clusters.

9.0 References

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10.0 Ethical and responsible AI considerations

This study uses the publicly available Home Credit Default dataset, which is fully anonymised, and no attempt is made to re-identify individuals. Data is stored securely on OneDrive but also on GitHub, which is not a controlled research environment. This represents a minor ethical limitation.

Generation of synthetic data using CTGAN may amplify any bias present in the original dataset, which is a known risk of GAN-based augmentation. To mitigate this, the quality of synthetic samples is evaluated using correlations between the distributional statistics of the synthetic and training data. This helps ensure that the characteristics of the augmented data remain consistent with the real population, to the extent possible within the methodological framework. Although misclassification costs in many class-imbalanced applications are frequently high, the classifiers trained in this study are used solely to generate empirical results and are not deployed to any production environment, thereby presenting no practical risk. The study also aligns with responsible AI principles by maintaining transparency in synthetic-data generation and evaluating the impact of class imbalance.

11.0 Appendix A: CTGAN hyperparameters and evaluation

Table 10. Hyperparameters values selected for CTGAN training. Only values that impact training are listed.

Hyperparameter	Selected Value
enforce_min_max_values	true
enforce_rounding	true
embedding_dim	64
generator_dim	[128, [128]
discriminator_dim	[128, 128]
generator_lr	0.0001
generator_decay	1e-06
discriminator_lr	0.0003
discriminator_decay	1e-06
batch_size	120
log_frequency	true
verbose	true
epochs	400
pac	10

Table 11. Kolmogorov-Smirnov statistic for numerical column for each sample size.

Column	Sample Size			
	545	2,234	5,586	22,342
AMT_REQ_CREDIT_BUREAU_WEEK	0.007	0.006	0.022	0.004
AMT_REQ_CREDIT_BUREAU_HOUR	0.002	0.003	0.002	0.005
AMT_REQ_CREDIT_BUREAU_DAY	0.006	0.004	0.001	0.006
DEF_60_CNT_SOCIAL_CIRCLE	0.011	0.012	0.015	0.008
ELEVATORS_MEDI	0.106	0.046	0.051	0.014
OWN_CAR_AGE	0.218	0.050	0.020	0.020
AMT_REQ_CREDIT_BUREAU_MON	0.083	0.059	0.003	0.021
OBS_30_CNT_SOCIAL_CIRCLE	0.128	0.040	0.097	0.031
OBS_60_CNT_SOCIAL_CIRCLE	0.090	0.155	0.030	0.032
HOUR_APPR_PROCESS_START	0.066	0.097	0.072	0.046
CNT_CHILDREN	0.075	0.050	0.025	0.050
DEF_30_CNT_SOCIAL_CIRCLE	0.022	0.021	0.074	0.053
DAYS_BIRTH	0.127	0.092	0.038	0.056
AMT_REQ_CREDIT_BUREAU_QRT	0.037	0.045	0.011	0.060
CNT_FAM_MEMBERS	0.070	0.057	0.095	0.062
AMT_CREDIT	0.088	0.148	0.092	0.075
DAYS_ID_PUBLISH	0.048	0.106	0.119	0.079
DAYS_LAST_PHONE_CHANGE	0.105	0.088	0.080	0.092
EXT_SOURCE_2	0.077	0.039	0.121	0.100
AMT_GOODS_PRICE	0.167	0.152	0.078	0.104
AMT_ANNUITY	0.040	0.086	0.168	0.114
REGION_POPULATION_RELATIVE	0.099	0.107	0.099	0.142
AMT_REQ_CREDIT_BUREAU_YEAR	0.029	0.043	0.021	0.149
DAYS_REGISTRATION	0.075	0.111	0.110	0.158
EXT_SOURCE_3	0.145	0.149	0.239	0.218
YEARS_BEGINEXPLUATATION_AVG	0.420	0.281	0.418	0.280
AMT_INCOME_TOTAL	0.116	0.168	0.064	0.294
NONLIVINGAREA_MODE	0.387	0.509	0.449	0.312
LIVINGAREA_AVG	0.391	0.355	0.306	0.322
NONLIVINGAPARTMENTS_AVG	0.464	0.556	0.444	0.343
LIVINGAREA_MODE	0.294	0.408	0.318	0.354
ENTRANCES_MODE	0.394	0.380	0.443	0.368
BASEMENTAREA_MEDI	0.415	0.494	0.437	0.371
LIVINGAPARTMENTS_AVG	0.365	0.519	0.579	0.373
TOTALAREA_MODE	0.360	0.328	0.391	0.373
APARTMENTS_MEDI	0.328	0.307	0.389	0.389
ELEVATORS_MODE	0.406	0.543	0.574	0.395
COMMONAREA_MEDI	0.455	0.371	0.523	0.404
APARTMENTS_AVG	0.345	0.412	0.431	0.410
APARTMENTS_MODE	0.306	0.456	0.404	0.415
EXT_SOURCE_1	0.323	0.394	0.382	0.417
LIVINGAPARTMENTS_MEDI	0.372	0.445	0.483	0.419
FLOORSMAX_MEDI	0.433	0.461	0.431	0.421
YEARS_BUILD_MODE	0.398	0.397	0.408	0.431

Column	Sample Size			
	545	2,234	5,586	22,342
BASEMENTAREA_AVG	0.393	0.443	0.378	0.434
LANDAREA_AVG	0.332	0.425	0.507	0.438
FLOORSMAX_MODE	0.413	0.476	0.422	0.441
DAYS_EMPLOYED	0.435	0.263	0.271	0.446
BASEMENTAREA_MODE	0.334	0.338	0.408	0.449
FLOORSMAX_AVG	0.372	0.374	0.495	0.452
NONLIVINGAPARTMENTS_MODE	0.422	0.247	0.336	0.456
FLOORSMIN_MEDI	0.426	0.576	0.553	0.456
YEARS_BEGINEXPLUATATION_MEDI	0.284	0.504	0.284	0.462
YEARS_BEGINEXPLUATATION_MODE	0.424	0.505	0.351	0.473
YEARS_BUILD_AVG	0.367	0.386	0.449	0.474
ENTRANCES_AVG	0.387	0.453	0.450	0.474
LANDAREA_MEDI	0.428	0.348	0.433	0.475
LIVINGAREA_MEDI	0.339	0.417	0.341	0.475
FLOORSMIN_AVG	0.411	0.461	0.424	0.475
NONLIVINGAREA_AVG	0.310	0.358	0.479	0.479
ENTRANCES_MEDI	0.420	0.326	0.388	0.483
COMMONAREA_MODE	0.361	0.482	0.437	0.504
COMMONAREA_AVG	0.367	0.479	0.506	0.505
LANDAREA_MODE	0.371	0.403	0.338	0.506
FLOORSMIN_MODE	0.459	0.533	0.504	0.518
NONLIVINGAPARTMENTS_MEDI	0.503	0.353	0.653	0.547
NONLIVINGAREA_MEDI	0.330	0.310	0.394	0.555
LIVINGAPARTMENTS_MODE	0.470	0.425	0.429	0.586
YEARS_BUILD_MEDI	0.380	0.542	0.401	0.603
ELEVATORS_AVG	0.543	0.673	0.351	0.628

Table 12. Boolean probabilities for sample size 22,342. Synthetic data values are close to real data for all columns, with no instances of mode collapse.

Column	Difference	P(T)-Real	P(T)-Syn	Mode Collapse
NAME_CONTRACT_TYPE	0.010	0.935	0.946	False
FLAG_OWN_CAR	0.003	0.304	0.302	False
FLAG_OWN_REALTY	0.033	0.684	0.651	False
FLAG_MOBIL	0.000	1.000	1.000	False
FLAG_EMP_PHONE	0.005	0.880	0.875	False
FLAG_WORK_PHONE	0.239	0.239	0.478	False
FLAG_CONT_MOBILE	0.006	0.998	0.992	False
FLAG_PHONE	0.016	0.246	0.262	False
FLAG_EMAIL	0.004	0.056	0.061	False
REG_REGION_NOT_LIVE_REGION	0.012	0.017	0.030	False
REG_REGION_NOT_WORK_REGION	0.023	0.056	0.079	False
LIVE_REGION_NOT_WORK_REGION	0.046	0.043	0.089	False
REG_CITY_NOT_LIVE_CITY	0.067	0.119	0.186	False
REG_CITY_NOT_WORK_CITY	0.007	0.303	0.310	False
LIVE_CITY_NOT_WORK_CITY	0.006	0.221	0.227	False
FLAG_DOCUMENT_2	0.003	0.000	0.004	False
FLAG_DOCUMENT_3	0.051	0.778	0.727	False
FLAG_DOCUMENT_4	0.000	0.000	0.000	False
FLAG_DOCUMENT_5	0.009	0.015	0.025	False
FLAG_DOCUMENT_6	0.006	0.061	0.055	False
FLAG_DOCUMENT_7	0.003	0.000	0.003	False
FLAG_DOCUMENT_8	0.023	0.073	0.051	False
FLAG_DOCUMENT_9	0.006	0.003	0.009	False
FLAG_DOCUMENT_10	0.000	0.000	0.000	False
FLAG_DOCUMENT_11	0.005	0.003	0.007	False
FLAG_DOCUMENT_12	0.000	0.000	0.000	False
FLAG_DOCUMENT_13	0.005	0.001	0.007	False
FLAG_DOCUMENT_14	0.004	0.001	0.006	False
FLAG_DOCUMENT_15	0.003	0.000	0.004	False
FLAG_DOCUMENT_16	0.006	0.006	0.012	False
FLAG_DOCUMENT_17	0.003	0.000	0.003	False
FLAG_DOCUMENT_18	0.005	0.006	0.011	False
FLAG_DOCUMENT_19	0.005	0.000	0.005	False
FLAG_DOCUMENT_20	0.005	0.001	0.006	False
FLAG_DOCUMENT_21	0.005	0.001	0.005	False
AMT_REQ_CREDIT_BUREAU_MISSING	0.045	0.172	0.217	False
EXT_SOURCE_1_MISSING	0.094	0.594	0.688	False
EXT_SOURCE_3_MISSING	0.024	0.228	0.252	False
OWN_CAR_AGE_MISSING	0.019	0.696	0.715	False
STATS_MISSING	0.028	0.605	0.577	False

Table 13. Categorical fidelity for sample size 22,342 including Jenson-Shannon Divergence (JSD), Chi-square, Chi-square norm, and number of missing categories. CTGAT fidelity is reasonable, with no mode collapse for any feature at higher sample sizes.

Column	JSD	Chi-Sq Norm	Chi-Sq	N Cats	N Missing Cats	Missings Cats (%)
CODE_GENDER	0	29.531	59.062	2	0	0%
NAME_TYPE_SUITE	0	5.386	43.084	8	0	0%
NAME_INCOME_TYPE	0.004	160.494	962.966	6	0	0%
NAME_EDUCATION_TYPE	0.001	16.89	84.452	5	0	0%
NAME_FAMILY_STATUS	0.003	119.783	598.917	5	0	0%
NAME_HOUSING_TYPE	0.012	281.119	1686.711	6	0	0%
OCCUPATION_TYPE	0.007	61.74	1173.062	19	0	0%
REGION_RATING_CLIENT	0.002	95.738	287.214	3	0	0%
REGION_RATING_CLIENT_W_CITY	0.002	116.394	349.183	3	0	0%
WEEKDAY_APPR_PROCESS_START	0.008	216.22	1513.543	7	0	0%
ORGANIZATION_TYPE	0.003	10.394	602.863	58	0	0%
FONDKAPREMONT_MODE	0.002	57.546	287.731	5	0	0%
HOUSETYPE_MODE	0.001	44.515	178.059	4	0	0%
WALLSMATERIAL_MODE	0.001	28.576	228.609	8	0	0%
EMERGENCYSTATE_MODE	0.001	49.902	149.706	3	0	0%

Table 14. CTGAN training runtimes in minutes by tier and IR.

Size Tier	IR 3	IR 9	IR 27	IR 81	IR 243
Large	276.3	119.8	42.1	12.2	4.2
Medium	147.8	58.9	17.7	7.4	1.8
Small	73.1	31.5	8.3	3.0	2.2

Table 15. TRTR versus TSTR results show that the classifier did not learn to predict the positive class when using synthetic samples.

Metric	IR 81 Size: 44,684		IR 81 Size: 44,684		IR 81 Size: 44,684		IR 81 Size: 44,684	
	TRTR	TSTR	TRTR	TSTR	TRTR	TSTR	TRTR	TSTR
n_estimators	1200	1200	1200	1200	1200	1200	1200	1200
early_stopping_rounds	100	100	100	100	100	100	100	100
best_iteration	63	0	63	0	63	0	63	0
pr_auc	0.034	0.012	0.034	0.012	0.034	0.012	0.034	0.012
roc_auc	0.693	0.496	0.693	0.496	0.693	0.496	0.693	0.496
log_loss	0.065	0.142	0.065	0.142	0.065	0.142	0.065	0.142
max_prob	0.05	0.162	0.05	0.162	0.05	0.162	0.05	0.162
precision	0	0	0	0	0	0	0	0
recall	0	0	0	0	0	0	0	0
f1_score	0	0	0	0	0	0	0	0
best_threshold	0.017	0	0.017	0	0.017	0	0.017	0
precision_tuned	0.065	0.012	0.065	0.012	0.065	0.012	0.065	0.012
recall_tuned	0.165	1	0.165	1	0.165	1	0.165	1
f1_score_tuned	0.094	0.024	0.094	0.024	0.094	0.024	0.094	0.024

12.0 Appendix B: XGBoost hyperparameters

Table 16. The same patterns of hyperparameters were used for the smaller diagnostic scenarios as for the small scenarios in the main experimental grid.

scenario	colsample_bytree	learning_rate	reg_alpha	reg_lambda	subsample	scale_pos_weight	min_child_weight	max_depth	gamma
large-ir003	0.6	0.1	0.5	1	1	1.5	5	8	0
large-ir009	0.8	0.05	0	2	1	4.5	5	6	1
large-ir027	0.6	0.03	0.5	2	1	13.5	3	4	1
large-ir081	0.6	0.05	1	0.5	0.6	162	1	4	5
large-ir243	0.6	0.05	1	0.5	0.6	486	3	6	0
medium-ir001	0.6	0.1	1	2	0.8	0.5	5	8	5
medium-ir003	0.6	0.05	1	2	0.8	1.5	5	6	5
medium-ir009	0.6	0.05	0	2	1	4.5	5	6	5
medium-ir027	0.6	0.03	1	2	0.8	13.5	5	4	5
medium-ir081	0.6	0.05	0.5	2	1	40.5	3	4	0
medium-ir243	0.6	0.05	0.5	2	1	121.5	3	4	0
medium-ir729	0.6	0.05	0.5	2	1	364.5	3	4	0
small-ir001	0.6	0.05	0.5	2	0.8	0.5	5	6	1
small-ir003	0.6	0.05	1	2	0.8	1.5	3	4	5
small-ir009	0.6	0.05	0	2	1	4.5	5	4	5
small-ir027	0.8	0.03	0	2	0.8	13.5	1	4	0
small-ir081	0.6	0.1	1	0.5	0.8	81	3	4	5
small-ir243	0.6	0.1	1	0.5	0.8	243	3	4	5

13.0 Appendix C: Metric values versus IR

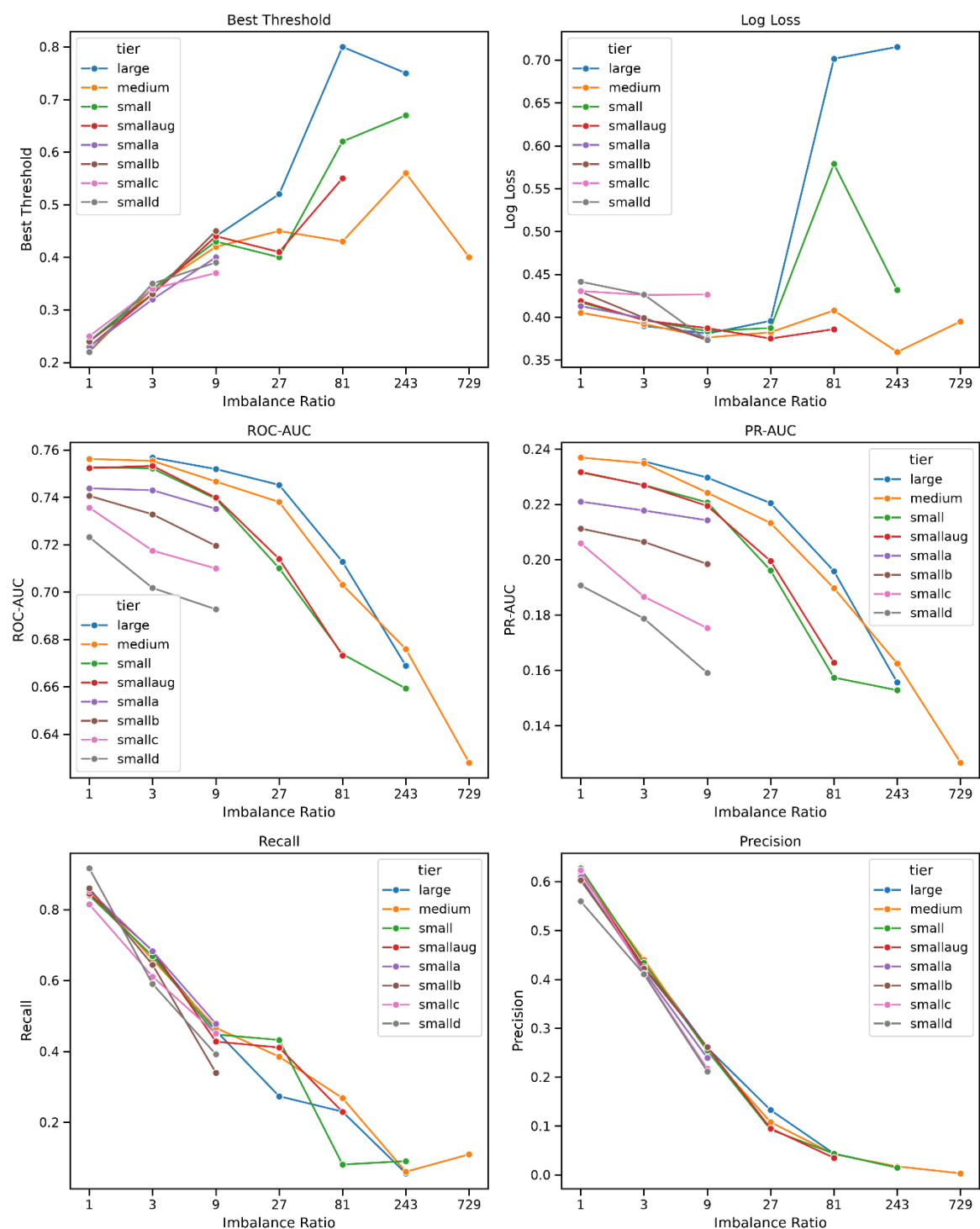


Figure 11. Best threshold, recall and precision for XGBoost vary approximately linearly with $\log(\text{IR})$. ROC-AUC and PR-AUC appear to have a more linear correlation to IR. While log-loss is unaffected until IR 81 then become unstable.

14.0 Appendix D: Effect of augmentation

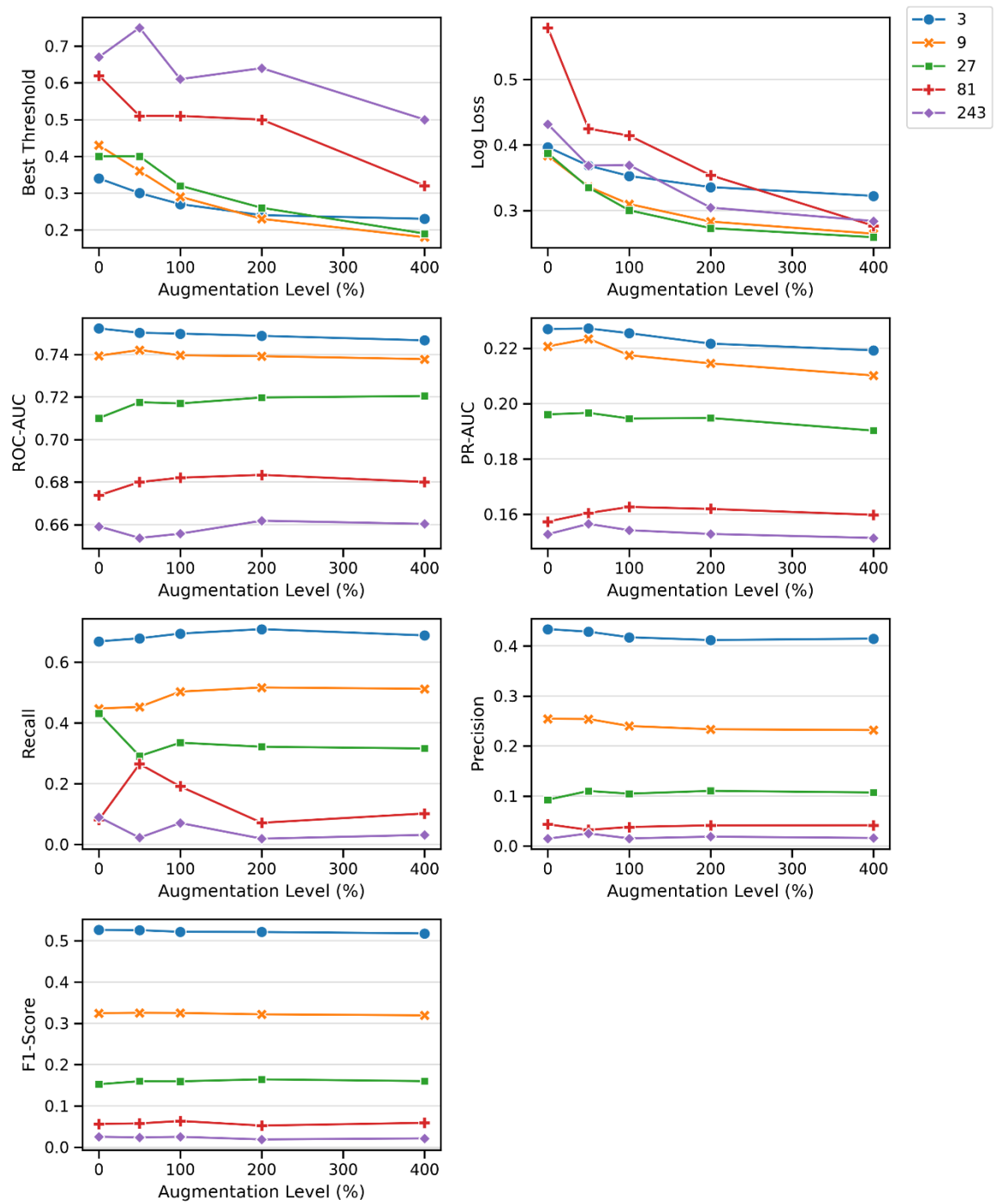


Figure 12. Small scenario augmentation effects for XGBoost. Best threshold and log-loss consistently fall with increasing augmentation, while other metrics show no discernible pattern.

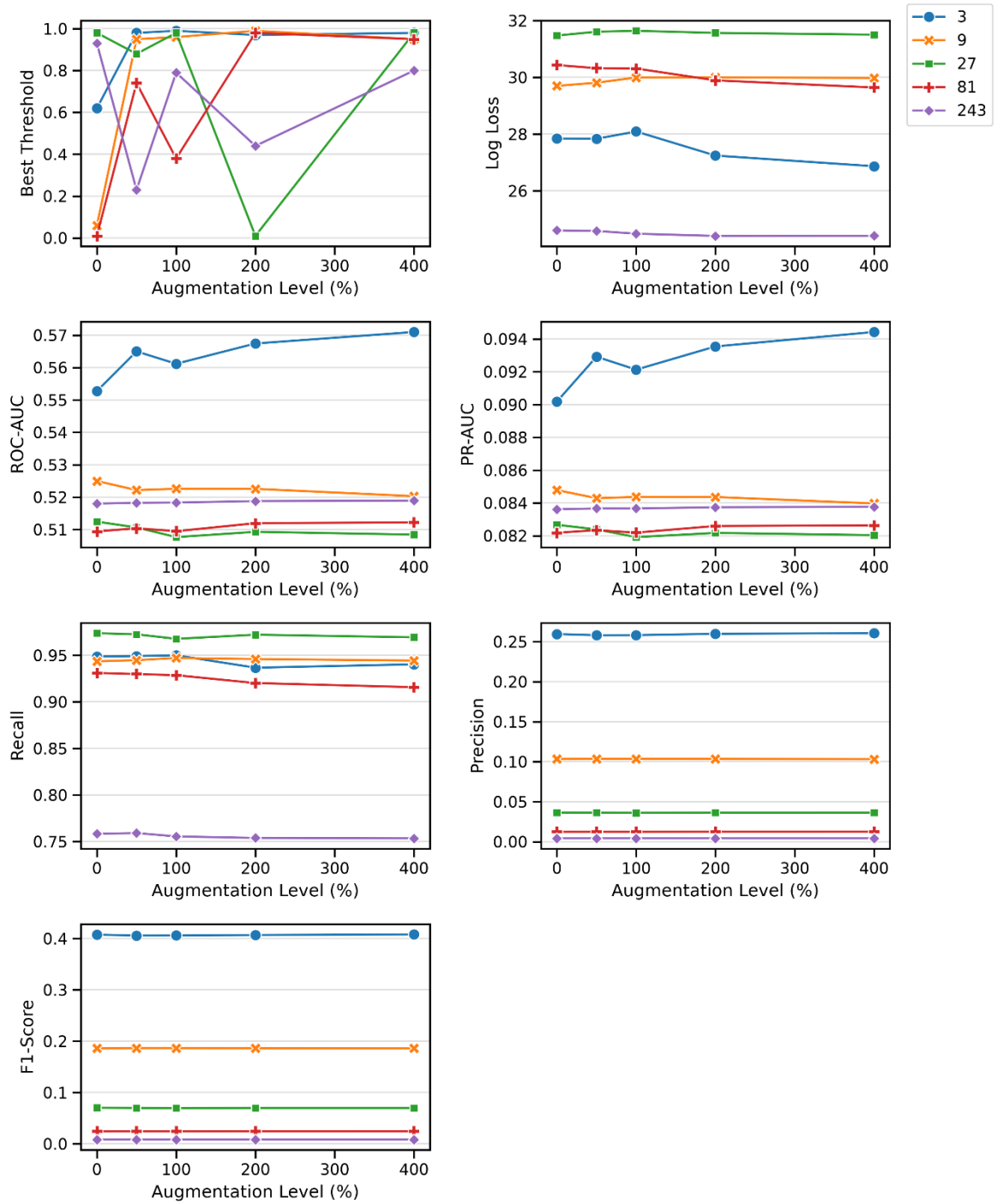


Figure 13. Small tier augmentation results for Gaussian naive Bayes. Threshold is not stable, but other metrics behave similarly to XGBoost.

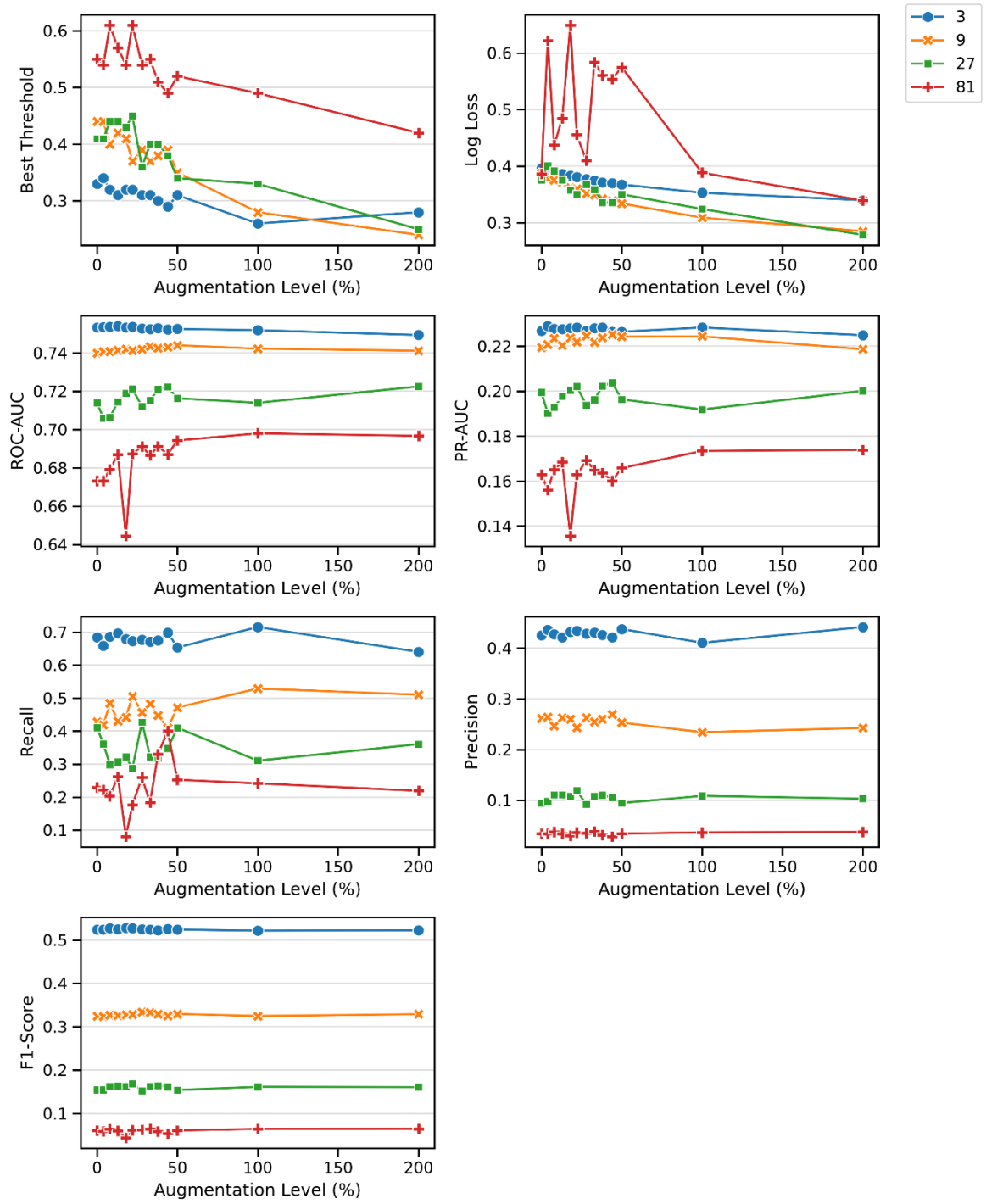


Figure 14. Augmenting the original small tier scenario with additional real observations produced similar results to the CTGAN synthetic data.

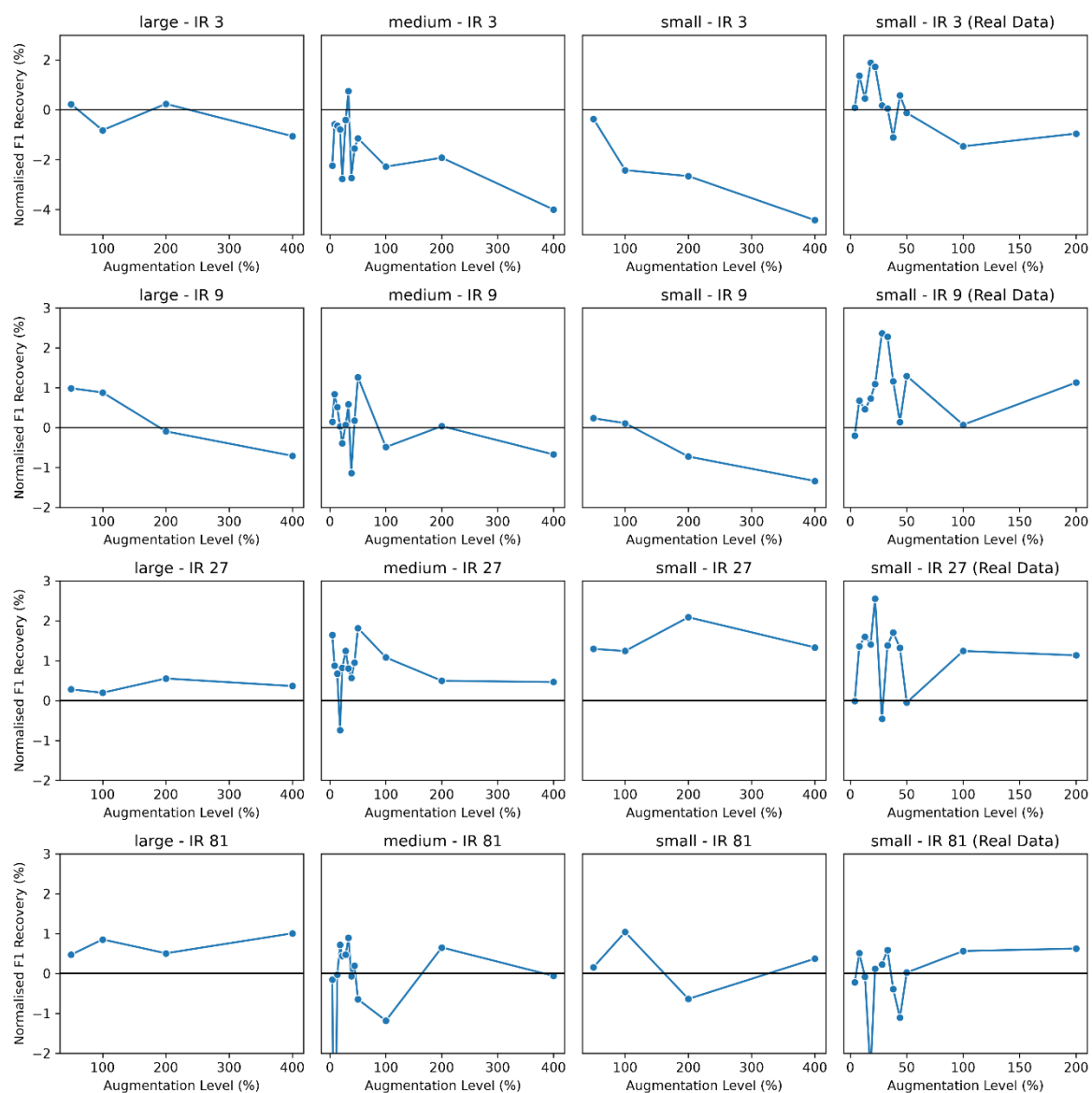


Figure 15. XGBoost Normalised F1 Recovery shows no consistent pattern of behaviour across IRs and sample sizes. The right-hand column shows results for augmentation with real data.

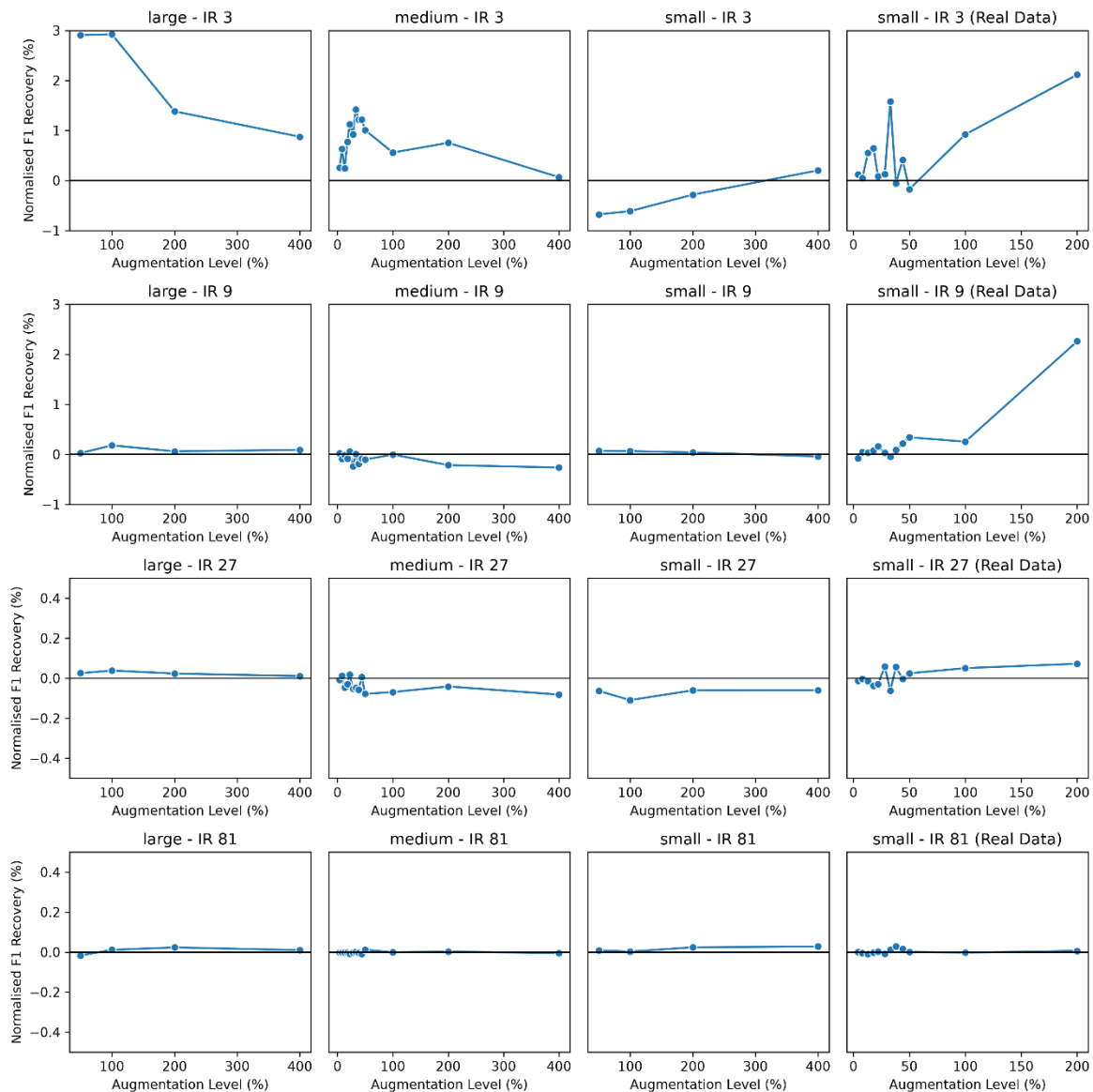


Figure 16. Gaussian naïve Bayes F1 Recovery percentage, as with XGBoost, shows consistent pattern of behaviour across IRs and sample sizes. The right-hand column shows results for augmentation with real data.

15.0 Appendix E: Development environment

The methodology is implemented in Python within JupyterLab and organised into a sequence of modular notebooks. This appendix provides a list of those notebooks, each corresponding to a distinct stage of the experimental pipeline from data preprocessing through to reporting. Notebooks themselves are available in the project repository:

<https://github.com/mendusahe/msc-dissertation.git>

01 – Data Preparation

- Convert features to the correct datatypes
- Replace missing values
- Create train and test split

02 – Experimental Scenarios

- Create sample indices for each scenario including cross-validation folds

03 – CTGAN Metadata Creation

- Define feature metadata required for CTGAN training

04 – CTGAN Data Synthesis

- Train CTGAN models for each scenario
- Generate synthetic data for each scenario and augmentation level

05 – CTGAN Evaluation

- Calculate fidelity metrics for numerical, categorical, and Boolean features
- Select hyperparameters
- Calculate CTGAN utility with TRTR versus TSTR

06 – XGBoost Hyperparameter Tuning

- Grid search for best hyperparameters for XGBoost

07 – XGBoost Hyperparameter Configuration

- Utility to read hyperparameters saved in a spreadsheet and write to JSON files

08 – Training and Evaluation

- Implements XGBoost and trains on each unaugmented and augmented scenario
- Cross-validation and threshold tuning
- Evaluation against the test set
- Gaussian Naïve Bayes training and evaluation
- Augmentation with real data
- F1 recovery metric calculation

09 – Reports

- Aggregation of CV and test set evaluation for XGBoost and Gaussian Naïve Bayes
- Visualisations of performance
- Simulation of balanced training set performing in imbalanced worlds
- Full metrics tabulation

16.0 Word count

Word counts	
Excluding headings, tables, figures, captions, and formulae.	
Introduction	484
Literature survey	1,309

Word counts Excluding headings, tables, figures, captions, and formulae.	
Data description	264
Proposed methodology	3,951
Results	1,315
Discussion	1,522
Conclusion	1,121
Total	9,966